

Unit 6

Timing and Critical Path Analysis

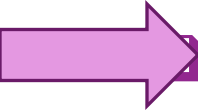
EL-GY 9463: INTRODUCTION TO HARDWARE DESIGN

PROFS. SUNDEEP RANGAN, SIDDHARTH GARG

Learning Objectives

- ❑ Write **SystemVerilog** descriptions for simple hardware **modules**
- ❑ Draw high-level **block diagram** representations of the module implementations
- ❑ Identify the **critical path** and its delay from the block diagram
- ❑ Implement simple functions in synchronous logic
- ❑ Visualize signals in a **timing diagram**
- ❑ Write a **testbench** for the hardware modules
- ❑ **Simulate**, and **synthesize** the SystemVerilog using **Vivado** tools
- ❑ Extract **value-change dump** (VCD) files and **synthesis reports** and process them in python

Outline

 Static Timing Analysis in Combinational Logic

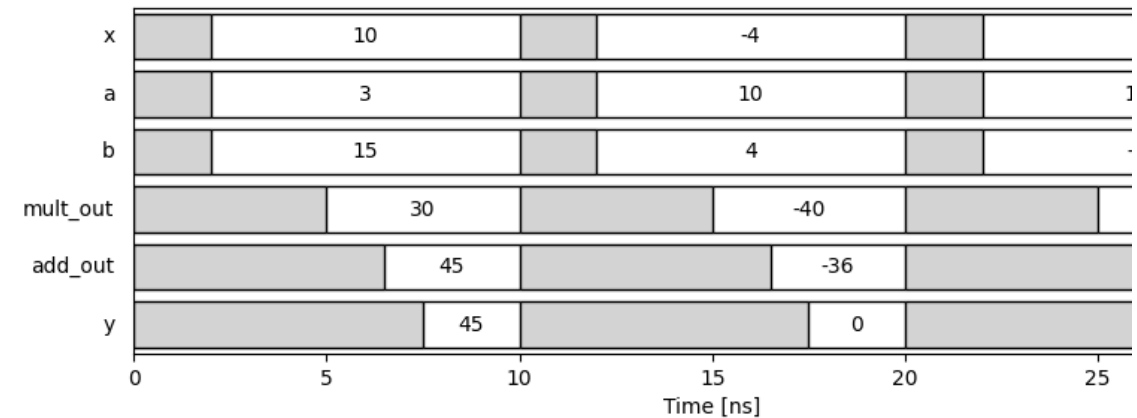
☐ Static Timing Analysis in Sequential Logic

☐ Sequential Logic Timing Diagrams

☐ VCD Files and Timing Diagrams from Vitis

Static Timing Analysis

- ❑ Timing analysis: Estimate delays through circuits
- ❑ Visualize with a **timing diagram**
- ❑ Critical for estimating:
 - Clock speed, latency, throughput, ...
- ❑ **Static timing analysis:**
 - Focus on **maximum propagation delay**
 - Timing that can be guaranteed for all inputs
 - Not data dependent
- ❑ This analysis is conservative, but provides hard guarantees
- ❑ Default analysis in industry



Implementing Signal Physically

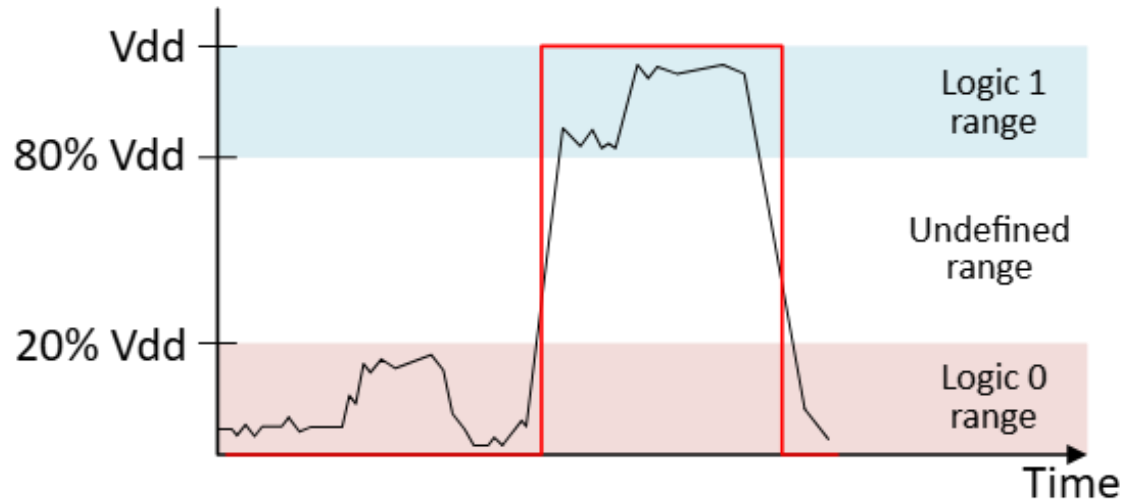
❑ To understand timing, we need to look at physical implementation of signals

❑ Consider a **binary signal** from Unit 1

❑ Two **logical values**: 0 or 1

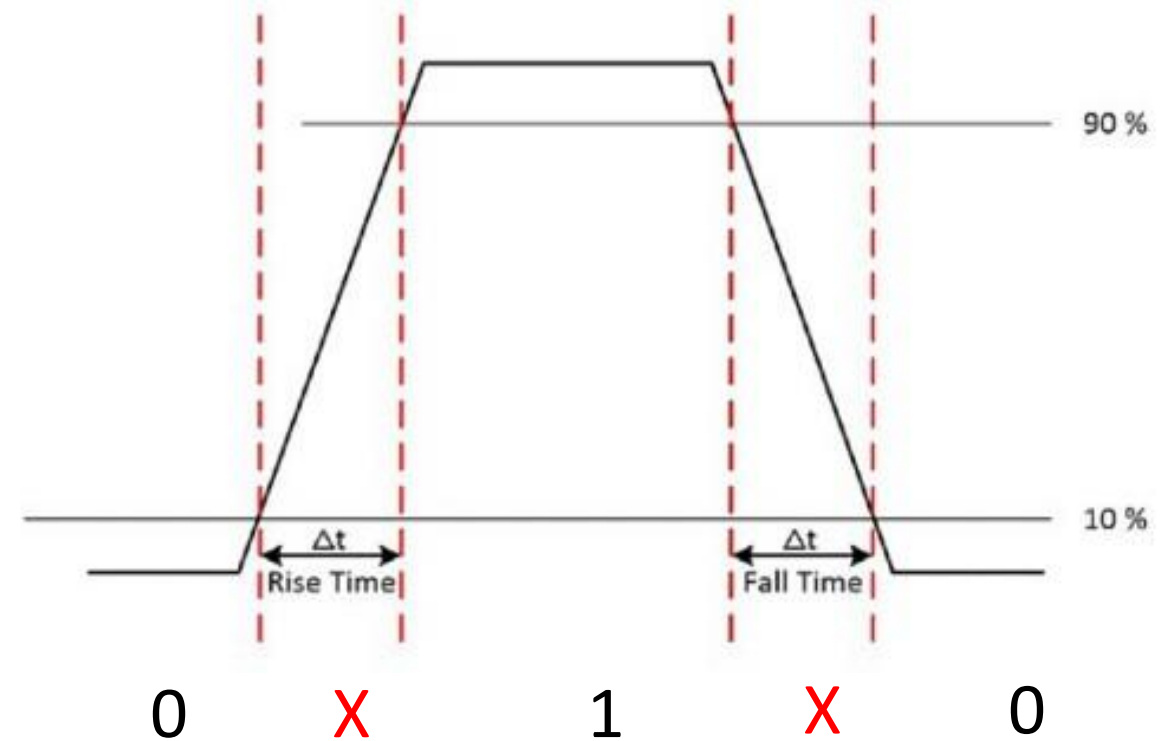
❑ What happens **physically**?

- Typically, a voltage on a wire
- Takes a continuous value
- Obtain logical value by thresholding

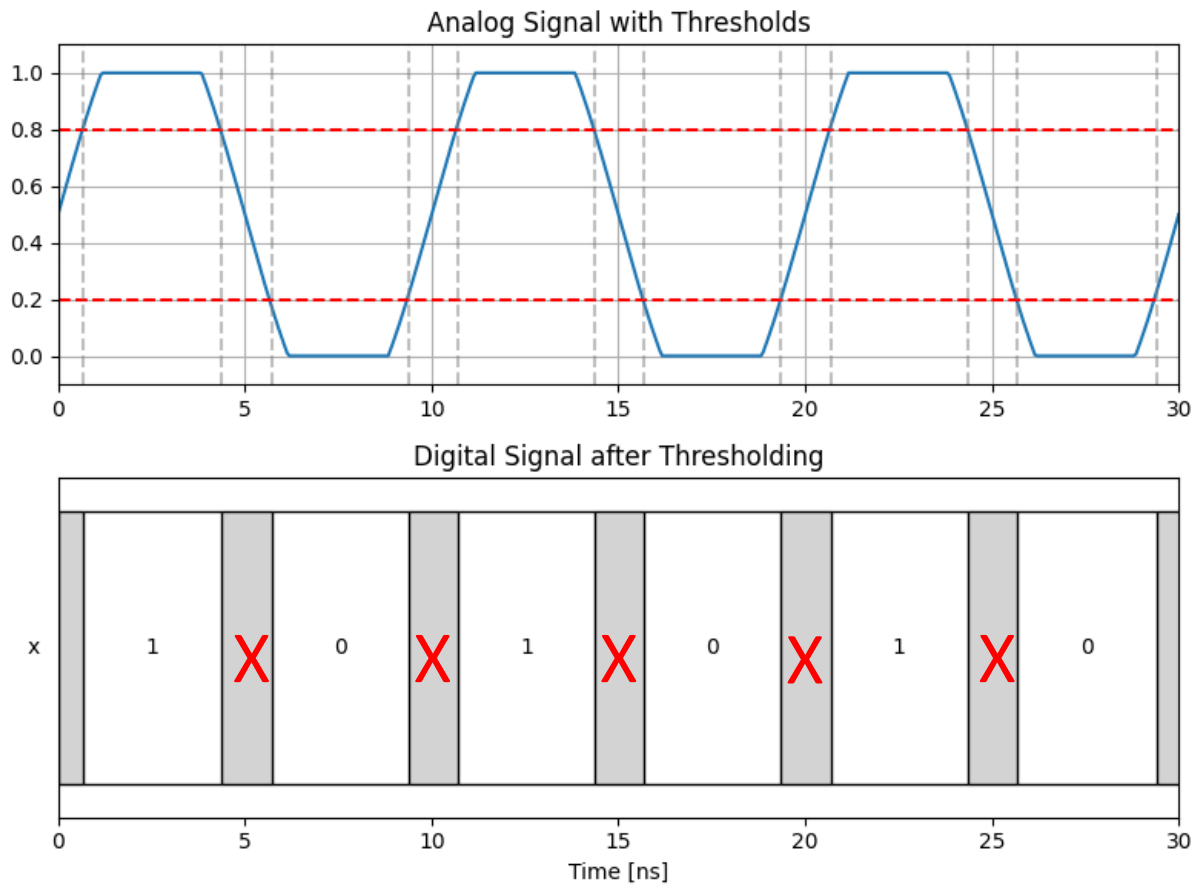


Undefined Logic Value: 'X' or '?'

- ❑ Signals take time to transition
 - They are outputs of physical circuits
 - Gates, computational units, ...
- ❑ Creates an **undefined** period
 - Also called indeterminate, unknown, ...
- ❑ Usually denoted as **X**
 - We may use **?** to avoid confusion with variable
- ❑ So binary logic signals have three values:
 - 0, 1 and X
 - Will discuss a fourth value Z later



Undefined Regions Illustrated



□ Analog physical signal

- Typically, a voltage
- Continuous valued
- Has finite transition periods

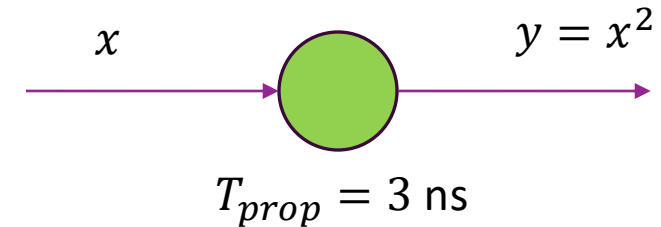
□ Digital logic signal

- Three values 0, 1 and X
- X value shown in gray

Stable Values and Propagation Delays

□ **Stable value:** Eventual value after a transition

- Ex: Suppose $y = x^2$ and x transitions from 4 to 5:
- Output y is eventually $y = 5^2 = 25$
- Stable output value is $y = 25$

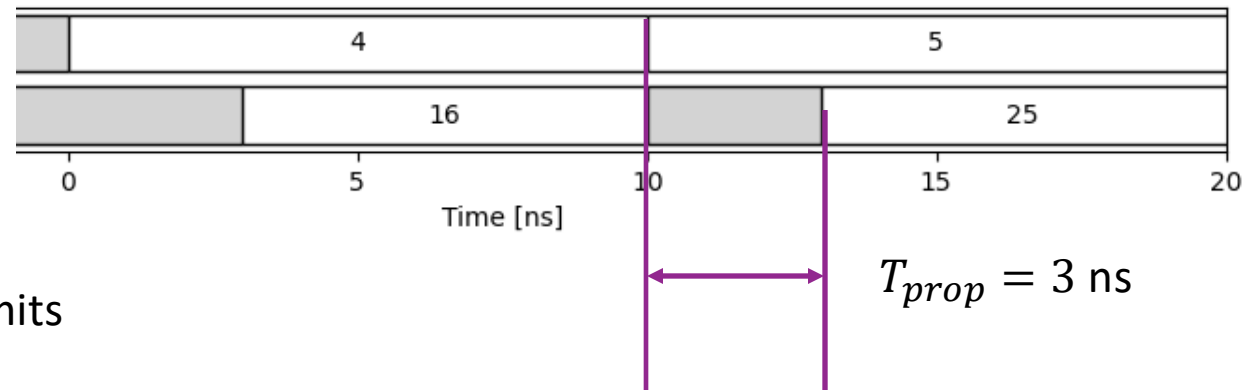


□ **Propagation delay:**

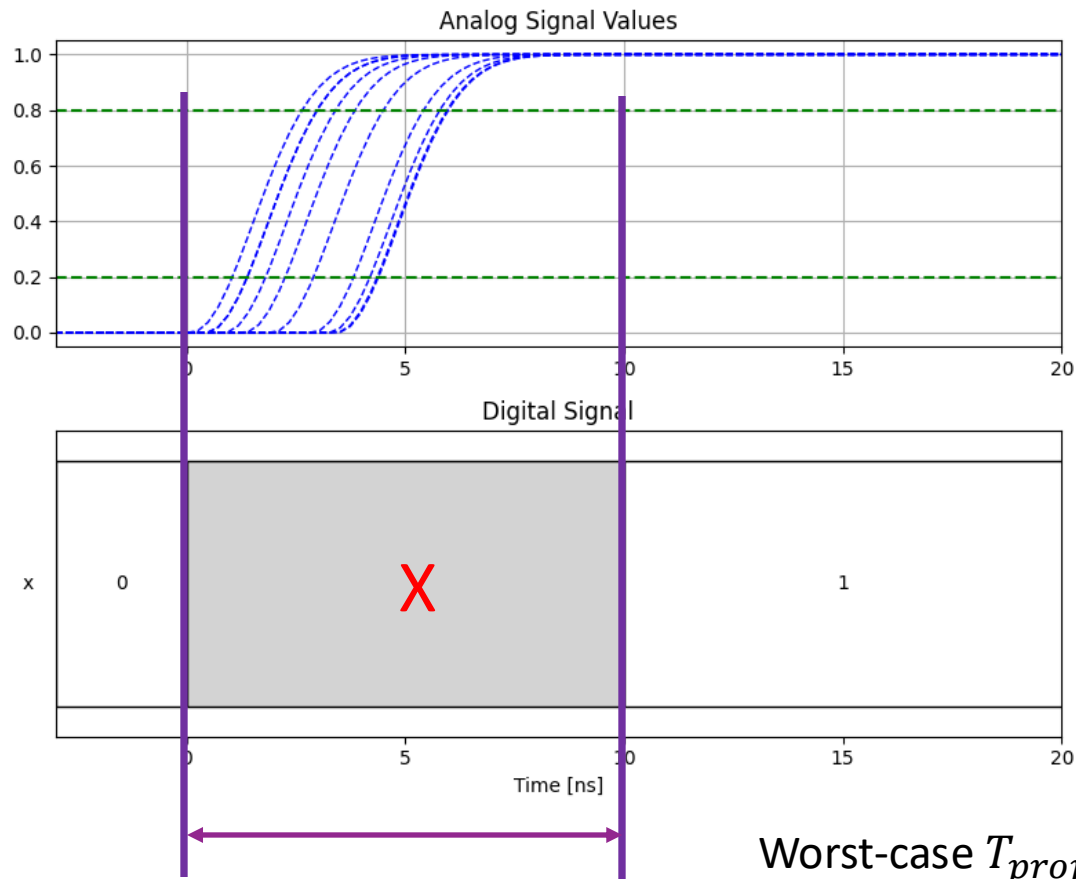
- Time for the output to transition

□ **Two components:**

- **Logic delay:** Time for the logic gates to settle
- **Routing delay:** Time to pass through wire
- May be related to capacitance, current source limits



Maximum Propagation Delay

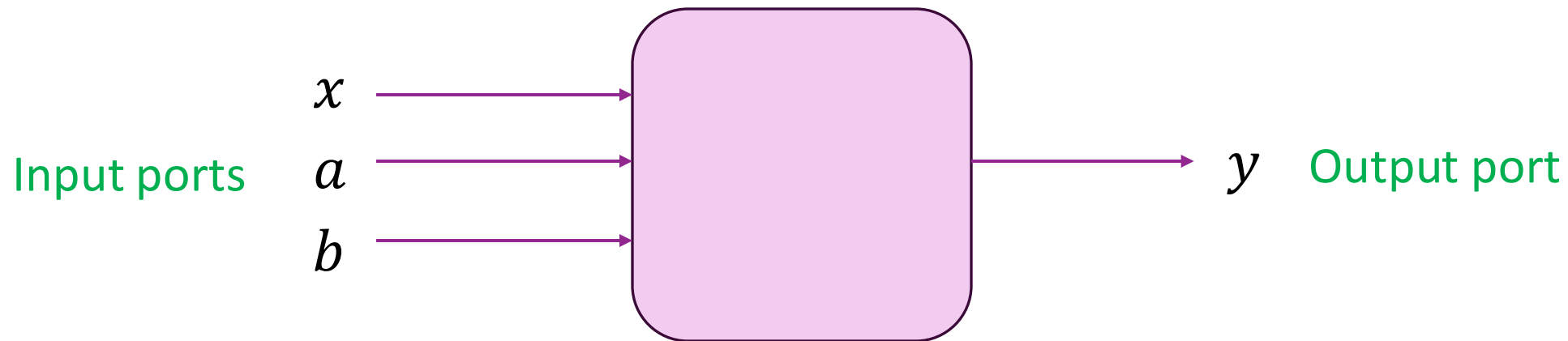


- ❑ Actual delays vary
 - Signal length, temperature, device variability
- ❑ Typically use maximum propagation delay
- ❑ Timing diagrams show max propagation delay
- ❑ X region represents:
 - Region where value cannot be guaranteed
 - Often stable before max delay
- ❑ Max propagation delay often very conservative
 - Real propagation delay may be $\ll$ maximum
 - Violating delay is usually catastrophic

Timing in a Series Circuit: ReLU + Linear

As example, consider a more complication circuit: $y = \max\{ax + b, 0\}$

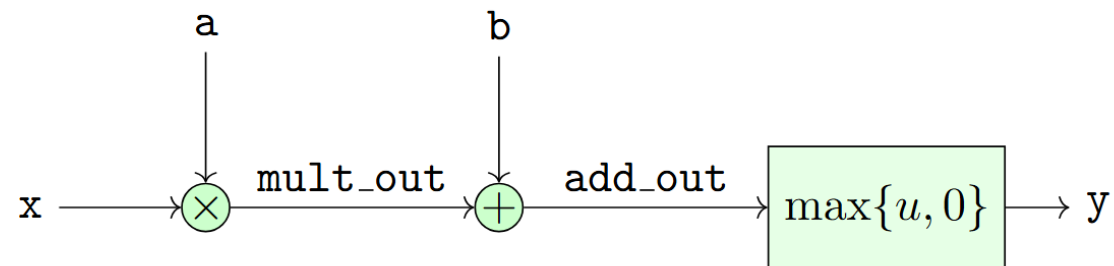
- Linear function + ReLU from Unit 2
- Used widely in machine learning



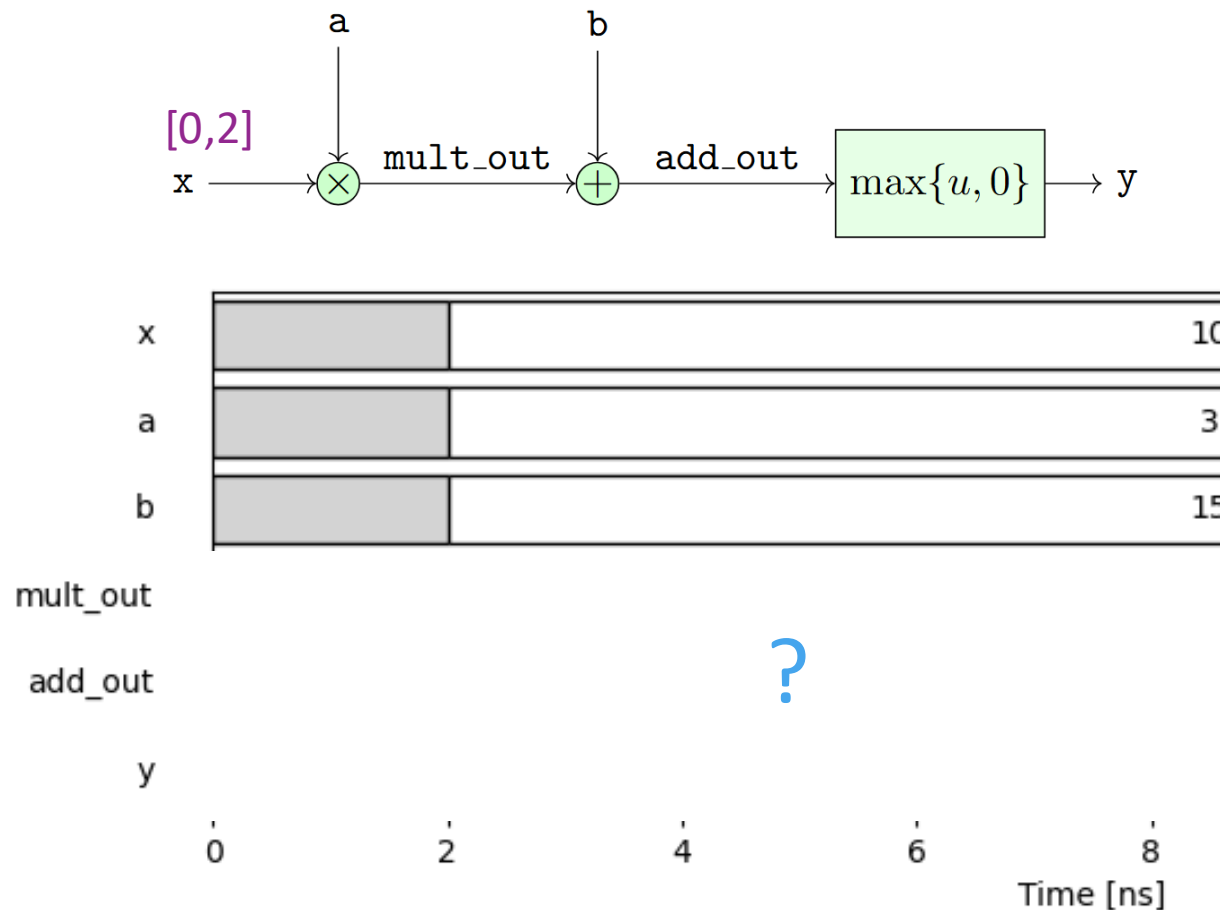
Timing Diagram for Series Connections

- ❑ Linear + ReLU can be implemented as a series connection
 - In this case, three operations one after the other
- ❑ For analysis, assume the max propagation delays in the table
- ❑ Question: What does the timing diagram look like with changes in the input?

Hardware block	Max prop delay
Multiplier	$T_{mult} = 3 \text{ ns}$
Adder	$T_{add} = 1.5 \text{ ns}$
Max operator	$T_{max} = 1 \text{ ns}$
Total	$T_{tot} = 5.5 \text{ ns}$



Timing Diagram of a Series



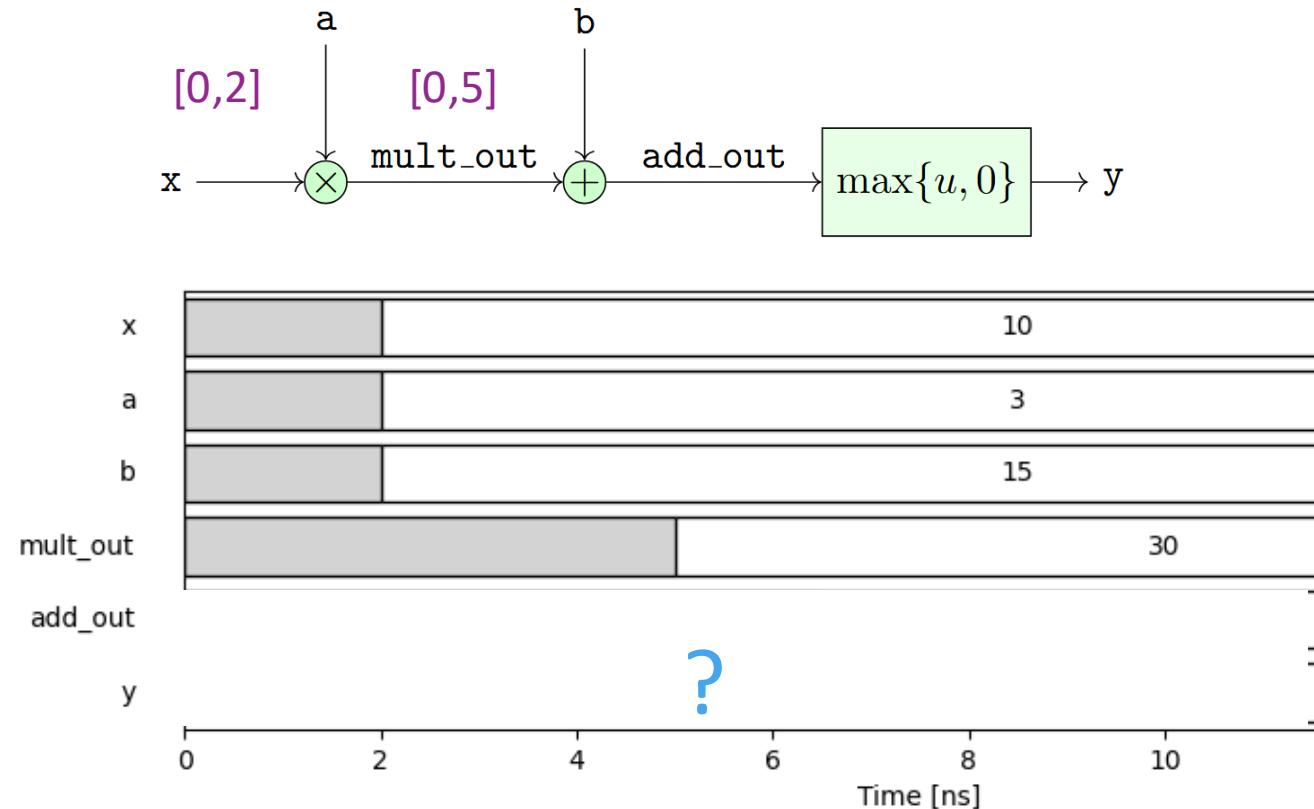
❑ Suppose x, a, b are invalid for $t \in [0,2]$

- They are in a transition in this period
- Signals are unknown X

❑ Mark the unknown region with violet $[0,2]$

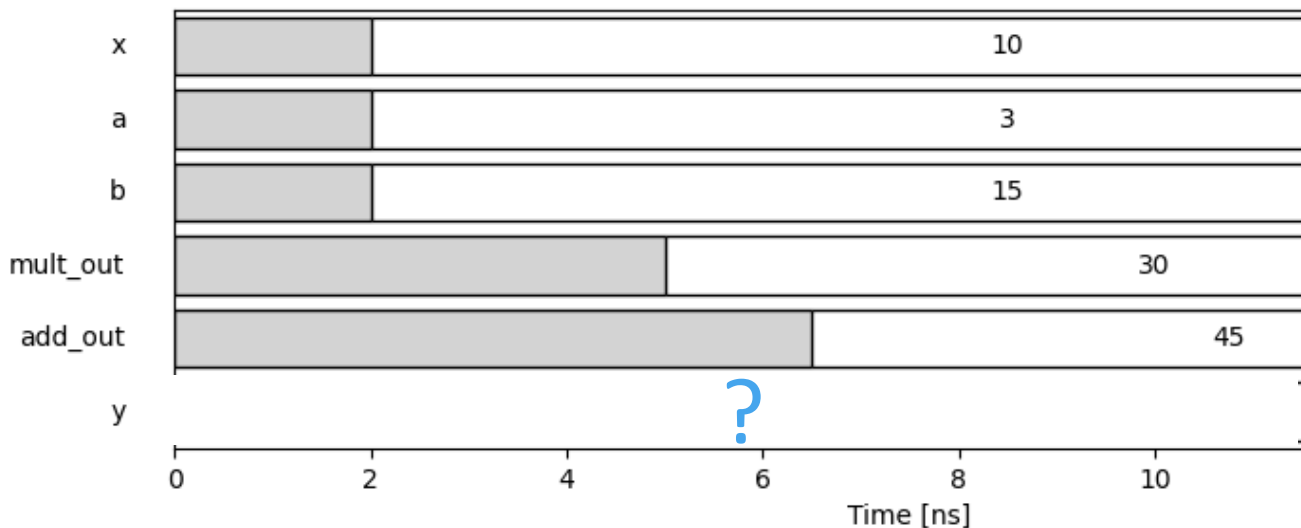
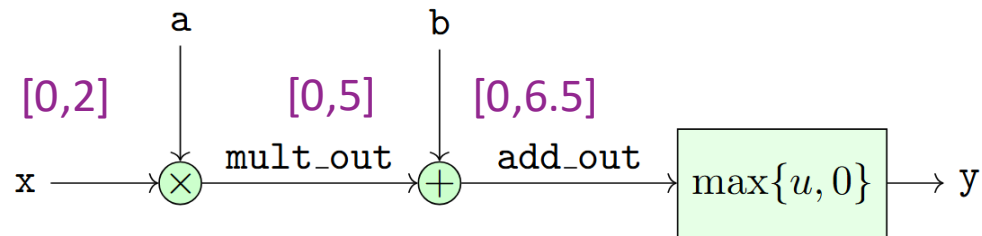
❑ When are downstream signals stable?

Series Timing: Initial Component



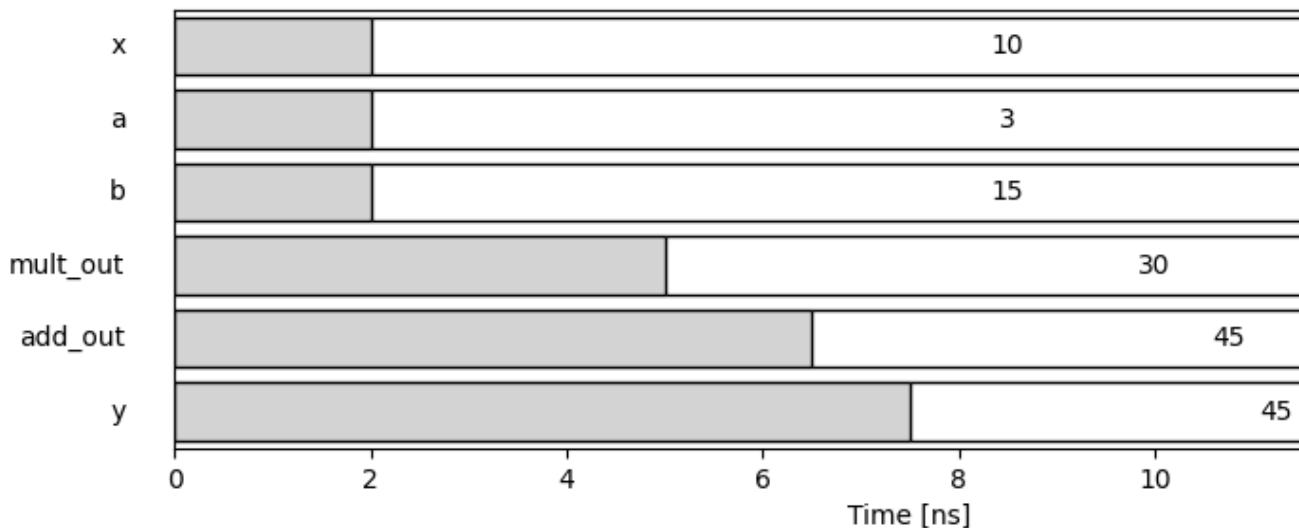
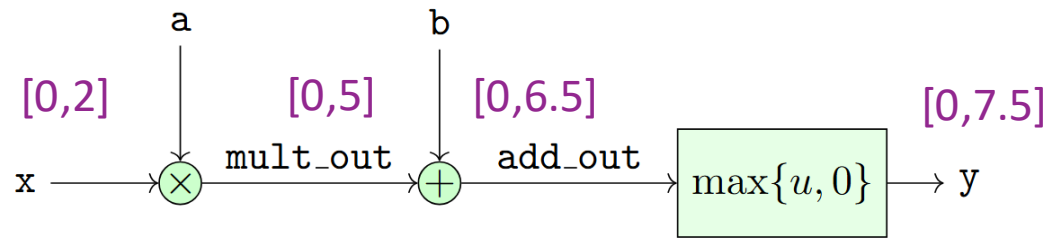
- ❑ Start with first component (multiplier)
- ❑ Inputs are invalid in [0,2]
- ❑ Max propagation delay of mult = 3 ns
 - Invalid for 3 ns longer than input
- ❑ Mult_out is invalid from [0,5]
 - Invalid for 3 ns longer than input
- ❑ Final stable value:
 - Mult_out = 3*10 = 30

Series Timing: Second Component



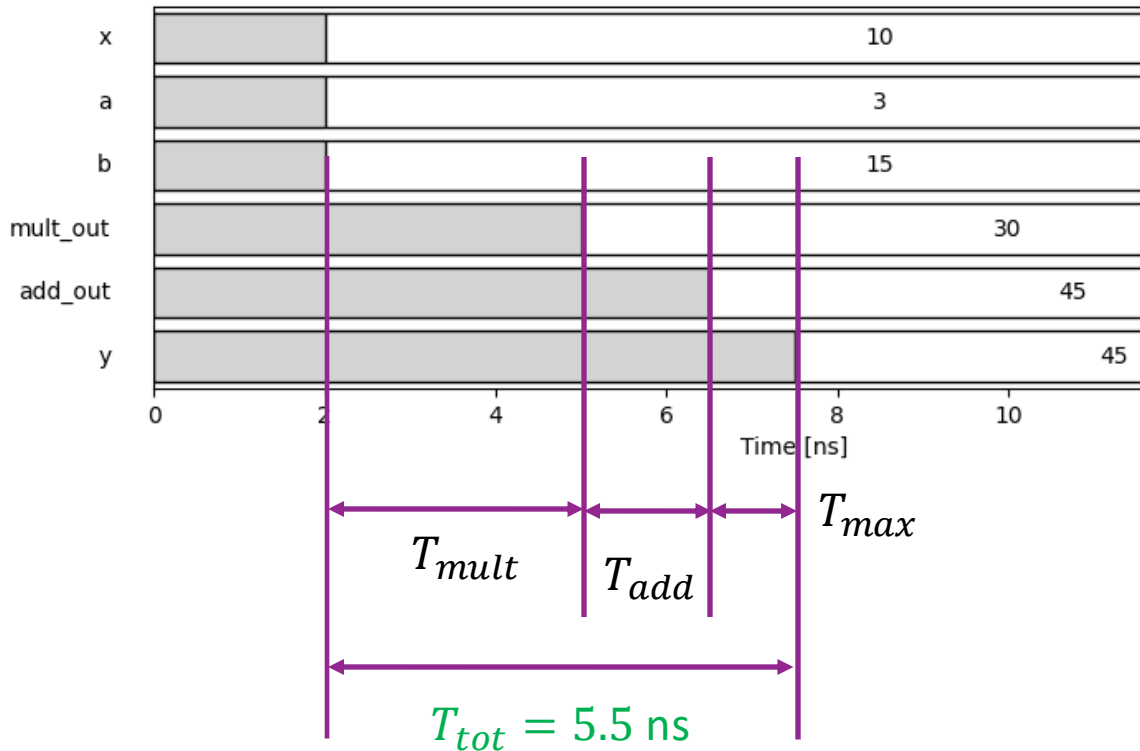
- ❑ From previous slide:
 - Mult_out is invalid from [0,5]
- ❑ Adder max propagation delay = 1.5 ns
- ❑ Hence add_out invalid in [0,6.5]
- ❑ Observe: For worst-case analysis
 - Add_out is invalid immediately when x is invalid
- ❑ Invalid propagates immediately
- ❑ Stable conditions propagate with delay

Series Timing: Third Component

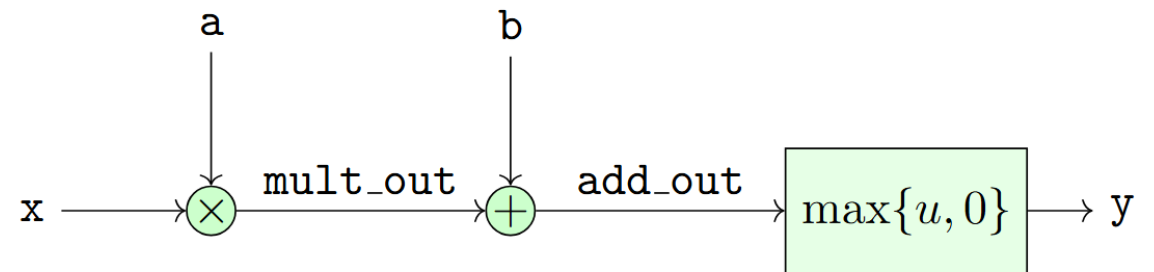


- ❑ Similar logic as `add_out`
- ❑ `Add_out` is invalid $[0, 6.5]$
- ❑ Max logic has max prop delay = 1 ns
- ❑ Output y is invalid in $[0, 7.5]$
- ❑ Total propagation delay = 5.5 ns
 - Period after inputs were stable

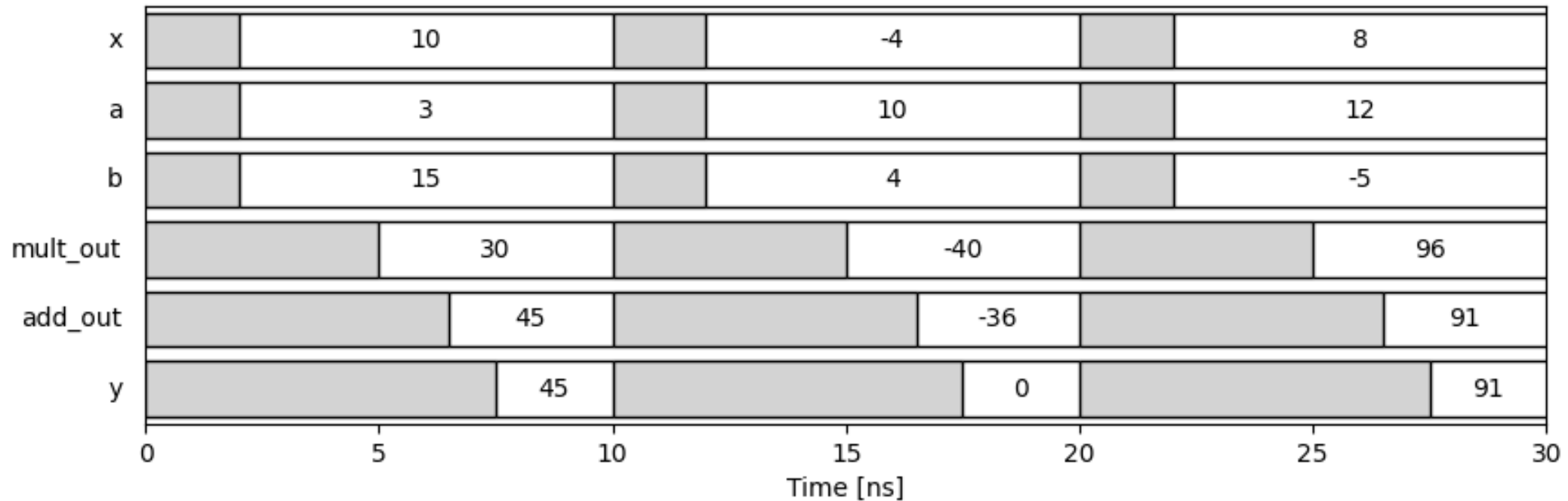
Example Total Propagation Delay



Hardware block	Example Value
Multiplier	$T_{mult} = 3$ ns
Adder	$T_{add} = 1.5$ ns
Max operator	$T_{max} = 1$ ns
Total	$T_{tot} = 5.5$ ns



Example Timing Over Multiple Inputs



Example Problem

Question 2. Series timing

Consider the following SystemVerilog combinational code:

```
xsq = x*x;  
y = a*xsq;  
z = y + b;
```

Assume each multiplication takes 3 ns and the addition takes 2 ns. Suppose that x is stable at $x=0$ for a long period before time 0, and then changes to $x=4$ at time 10 ns, remaining stable thereafter. The variables $a=3$ and $b=6$ are constants. meaning that x changes from unknown value at 0 ns to 4 at 10 ns.

- What are the stable values for xsq , y , and z before and after the change?
- What are the time-value change points of xsq , y , and z starting at time = 0 ns? Make sure to indicate the periods of the unknown values (X) during the propagation delays.

Typical Propagation Delays: Low-End FPGA

Hardware block	FPGA resource	Typical range
32-bit adder	LUT + dedicated carry chain	1.5–3.0 ns
16-bit multiplier	DSP48E1 block	2.5–5.0 ns
16-bit comparator	LUT logic + carry chain	1.0–2.0 ns

- ❑ Rough delays for a low-end FPGA (e.g., Pynq Z2)
- ❑ We will explain the FPGA resources more in later units
- ❑ Routing delays not included
 - May be large for more complex designs

Typical Propagation Delays: ASIC

Hardware block	Typical structure	Typical range
32-bit adder	Kogge-Stone / Brent-Kung / hybrid	150-450 ps
16-bit multiplier	Wallace/Dadda tree + final adder	300-700 ps
16-bit comparator	XOR/AND/OR chain with early-out	50-300 ps

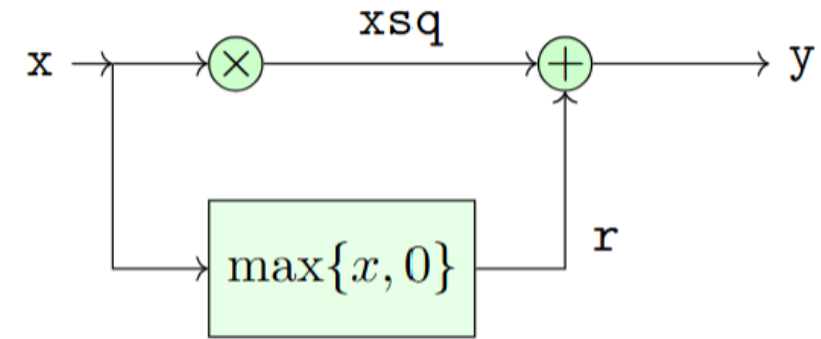
- ❑ Rough delays for TSMC 16nm
- ❑ Custom ASICs are much faster than FPGAs
 - Easily 5 to 10x faster

Timing Diagram with Multiple Paths

□ Now consider a circuit with multiple paths

□ Input to the adder has two paths:

- One from xsq
- Second from ReLU function r



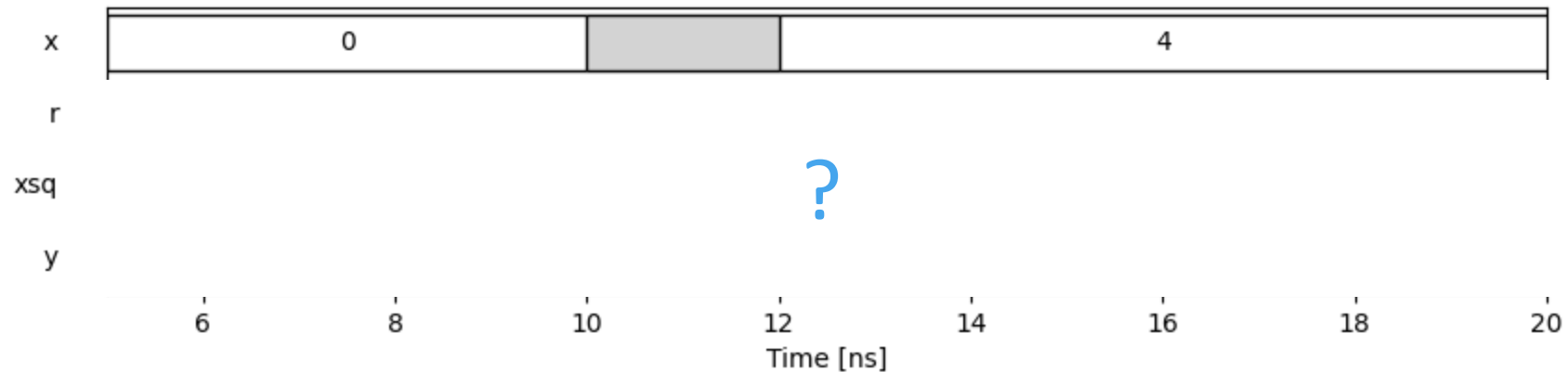
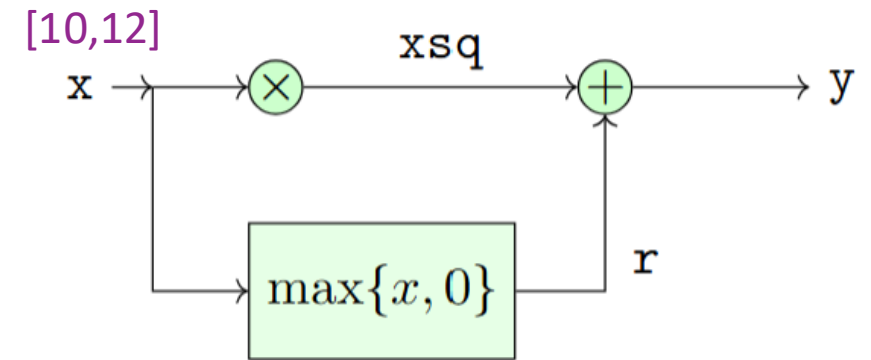
□ How does the timing diagram work?

```
xsq = x * x;  
r = (x > 0) ? x : 0;  
y = xsq + r;
```

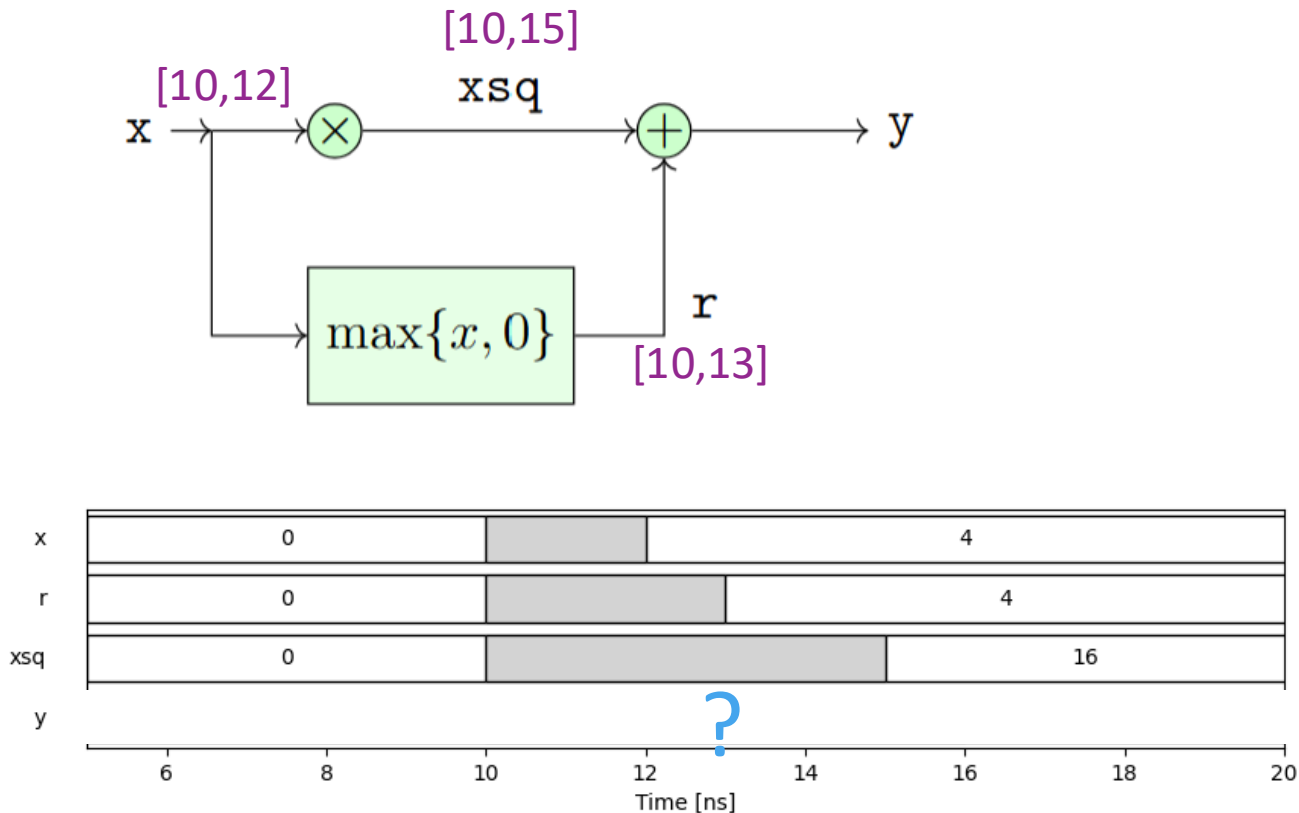
Timing Diagram: Inputs

□ Suppose input:

- Transitions from $x = 0$ to 4
- Transition interval is $t \in [10,12]$ (marked as X)



Timing Diagram: First Steps

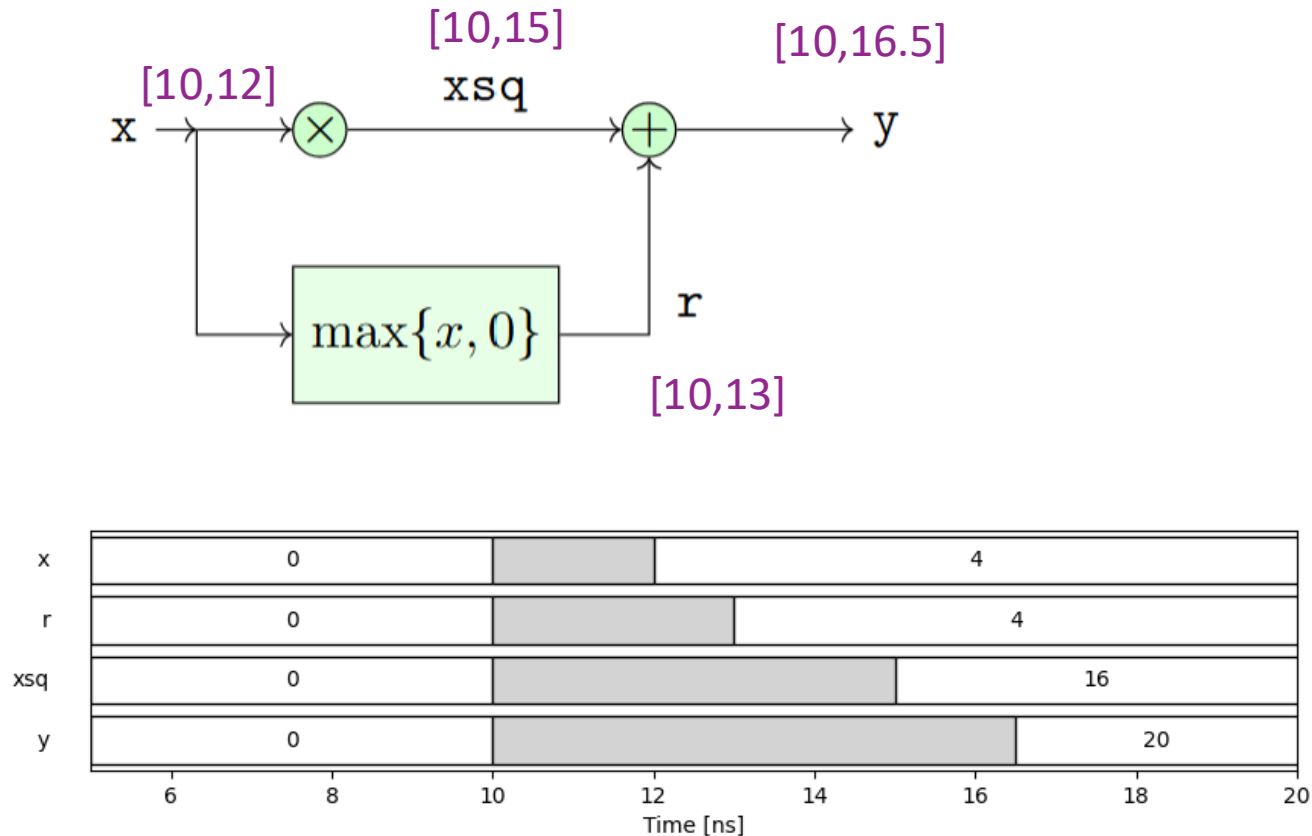


- Square output x^2
 - Multiplication max delay is 3
 - Value X from $[10, 15]$
 - Then comes to stable value 16

- ReLU output r
 - ReLU max delay is 1
 - Value X from $[10, 13]$
 - Then comes to stable value 4

- These steps operate in **parallel**

Multipath: Final Output



□ Output y has two inputs

- x^2 and r

□ Output goes invalid when either input = X

- Both are invalid starting at $t = 10$
- So, y is invalid starting at $t = 10$

□ Propagation delay starts when both are stable

- Before this output cannot be guaranteed
- x^2 is stable at $t = 15$
- r is stable at $t = 13$
- Propagation for adder is 1.5
- y stable at 16.5

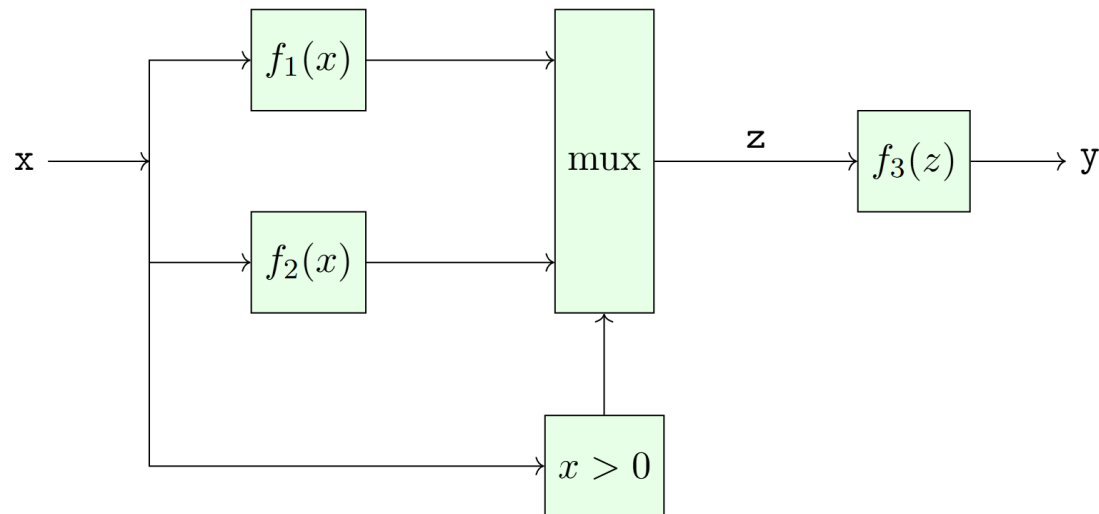
Conditional Circuits

- ❑ Consider conditional block
- ❑ How is it implemented?
- ❑ What is the max propagation delay?

```
if (x > 0) begin    // delay = 0.5
    z = func1(x);  // delay = 4
end else begin
    z = func2(x);  // delay = 2
end
y = func3(z);      // delay = 1.5
```

Implementing Conditional with Multiplexers

```
// mux delay = 1 ns
if (x > 0) begin // delay = 0.5
    z = func1(x); // delay = 4
end else begin
    z = func2(x); // delay = 2
end
y = func3(z); // delay = 1.5
```



❑ Multiplexer:

- Circuit that selects one of a set of inputs
- Uses a condition input

❑ Inputs and conditions

- All computed in parallel
- But each need distinct hardware

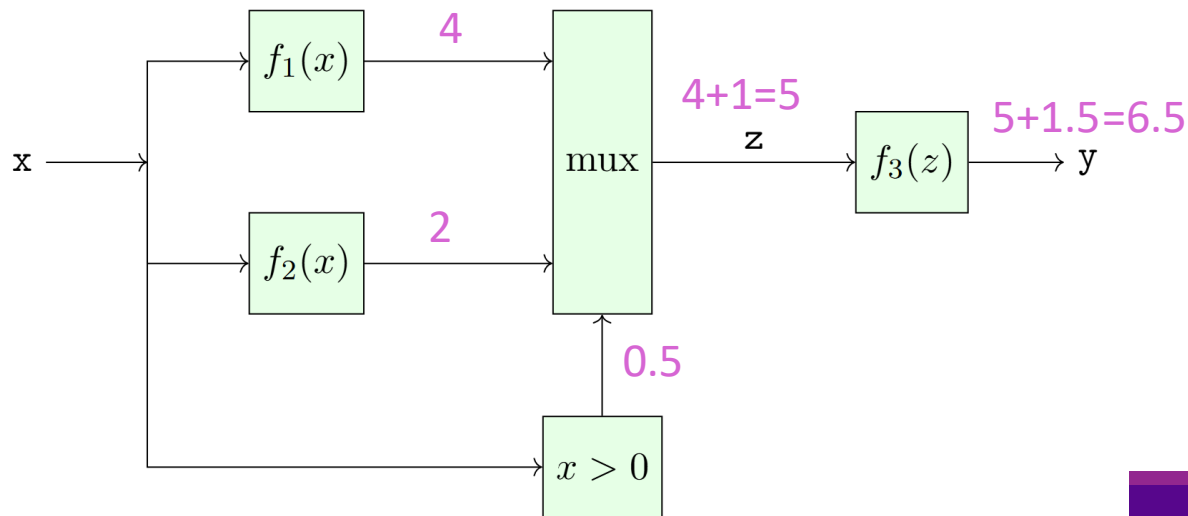
❑ No savings in hardware dependent on data

- All possible inputs must be computed

❑ Propagation delay starts with slowest input

Propagation Delay with Multiplexers

```
// mux delay = 1 ns
if (x > 0) begin // delay = 0.5
    z = func1(x); // delay = 4
end else begin
    z = func2(x); // delay = 2
end
y = func3(z); // delay = 1.5
```



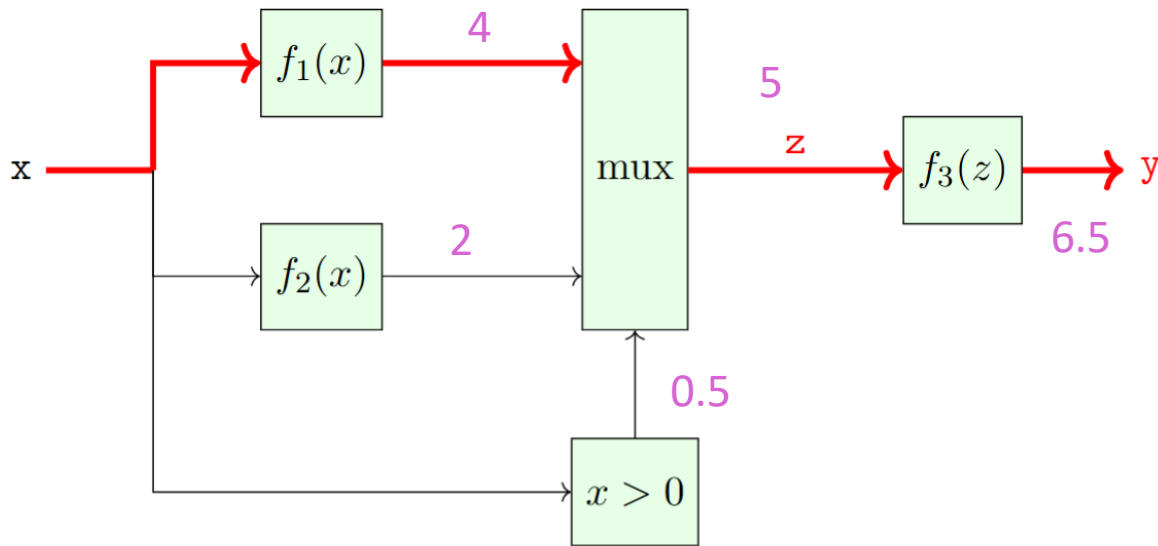
❑ Values in **violet** are max prop delay

❑ Slowest of the mux input and condition:

- $z = f_1(x)$ branch is the slowest
- Takes 4 ns
- Mux adds 1 ns
- So z is stable at 5 ns

❑ Output y is stable 1.5ns after z

Critical Path



❑ Critical path:

- The path with the largest max propagation delay

❑ In this case: $x \rightarrow f_1(x) \rightarrow z \rightarrow y$

❑ To reduce overall max propagation delay:

- Must reduce max delay on critical path

❑ For example:

- Reducing delay of $f_1(x)$ will reduce critical path
- But reducing $f_2(x)$ does not help

Critical Path Example

□ Example: For some functions $F_1, \dots, F_4$

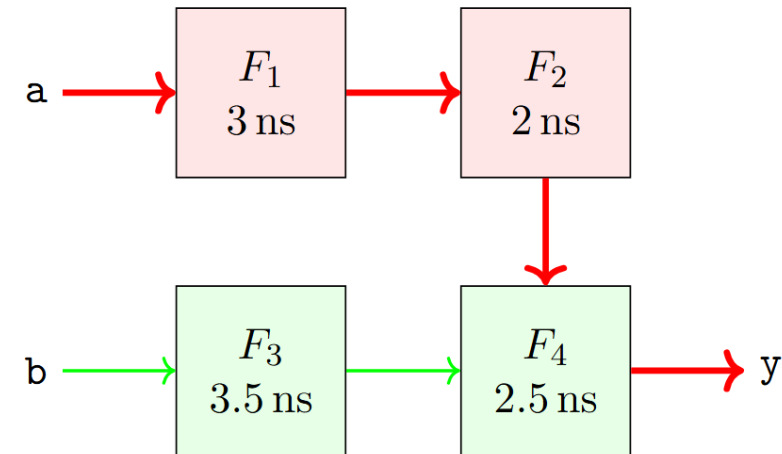
- $x_1 = F_2(F_1(a))$
- $x_2 = F_3(b)$
- $y = F_4(x_1, x_2)$

□ Path 1 ($F_1 \rightarrow F_2 \rightarrow F_4$):

- Delay = $3 + 2 + 2.5 = 7.5$ ns
- Longest = Critical path

□ Path 2 ($F_3 + F_4$):

- Delay = $3.5 + 2.5 = 6$ ns
- Shorter = Non-critical

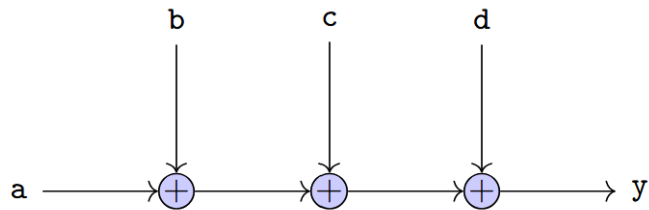


Delay Depends on Implementation

- Example: Consider $y = a + b + c + d$
- Suppose each add takes D seconds
- Synthesis tools try to find low delay implementations

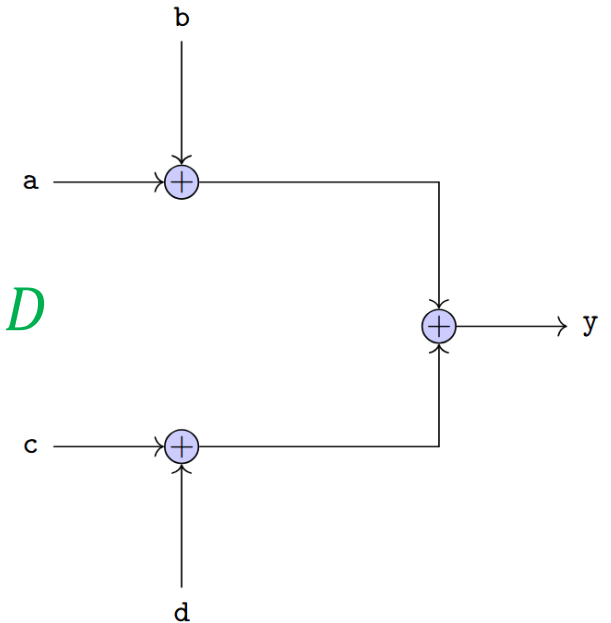
Serial implementation

Prop delay = $3D$



Parallel

Prop delay = $2D$



Outline

☐ Static Timing Analysis in Combinational Logic

 ☐ Static Timing Analysis in Sequential Logic

☐ Sequential Logic Timing Diagrams

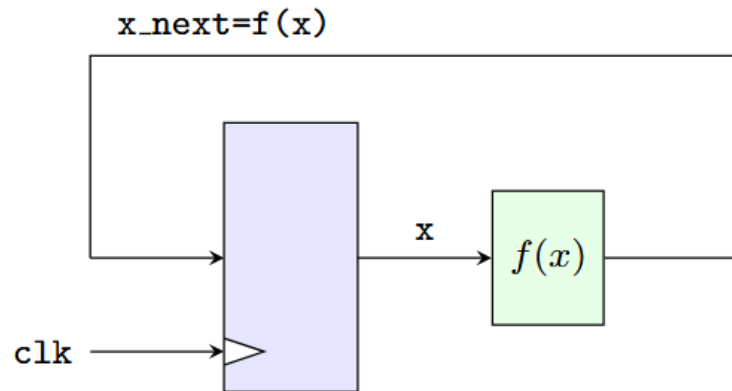
☐ VCD Files and Timing Diagrams from Vitis

Synchronous Logic

□ We now perform static timing analysis for synchronous logic

□ This unit, we assume

- All registers clocked on a common positive clock edge
- All registers are perfectly synchronized (no **clock skew**)
- No asynchronous reset



```
always_comb begin
    x_next = f(x);
end
always_ff @(posedge clk) begin
    x <= x_next;
end
```

STA for Synchronous Logic Overview

❑ STA considers all **register-to-register** paths

- Every register output to register input

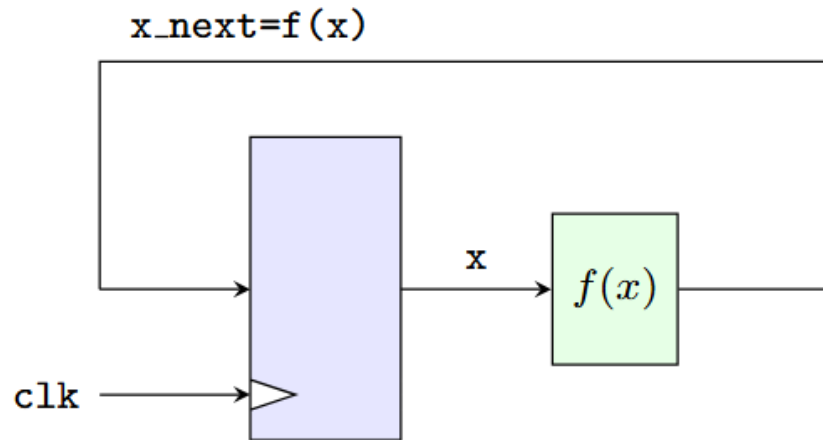
❑ Tests two conditions

❑ No paths arrive too late

- Signals must arrive before next transition
- Must meet a **set-up time** and **clock-to-Q time**

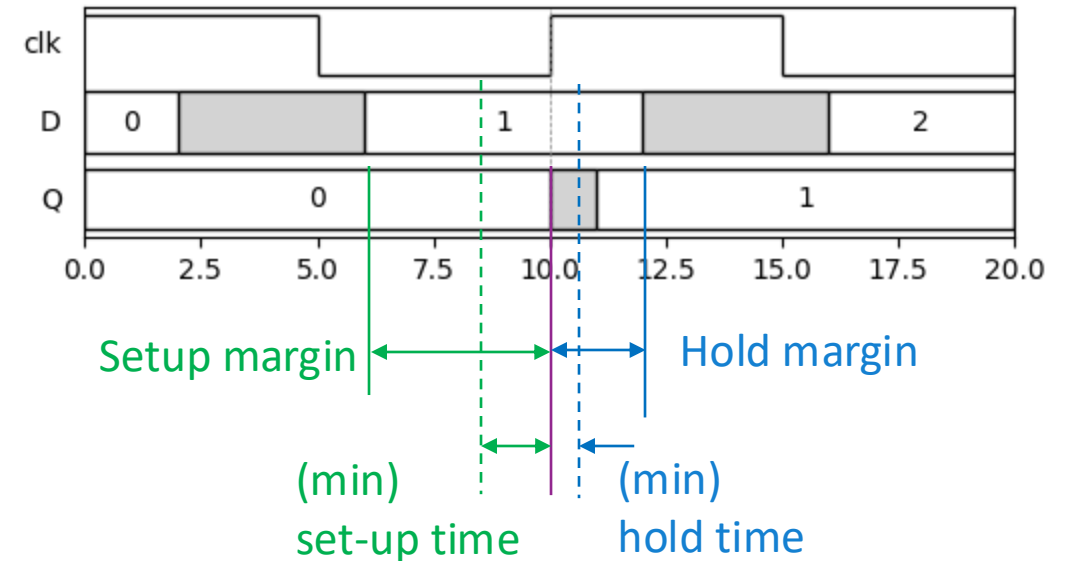
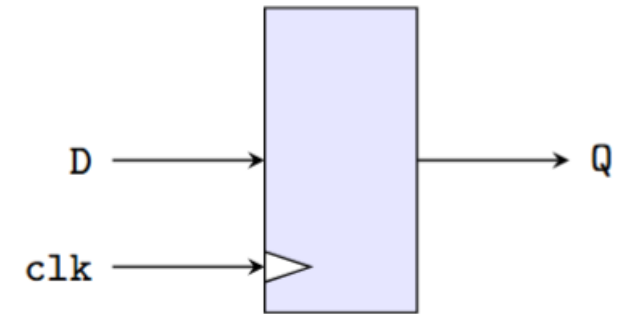
❑ No paths arrive too early

- Min delay > **hold time**
- Registers inputs must be stable for a hold period



Set-Up and Hold Time

- ❑ Registers typically implemented with a flip-flop
 - Sometimes called D-Q flip-flop
- ❑ Stable operation requires:
 - Inputs are stable before and after clock edge
- ❑ Set up margin:
 - Time from D stable to rising clock edge
 - Requirement: Set-up margin > set-up time
- ❑ Hold margin:
 - Time D stable after rising clock edge
 - Requirement: Hold margin > hold time



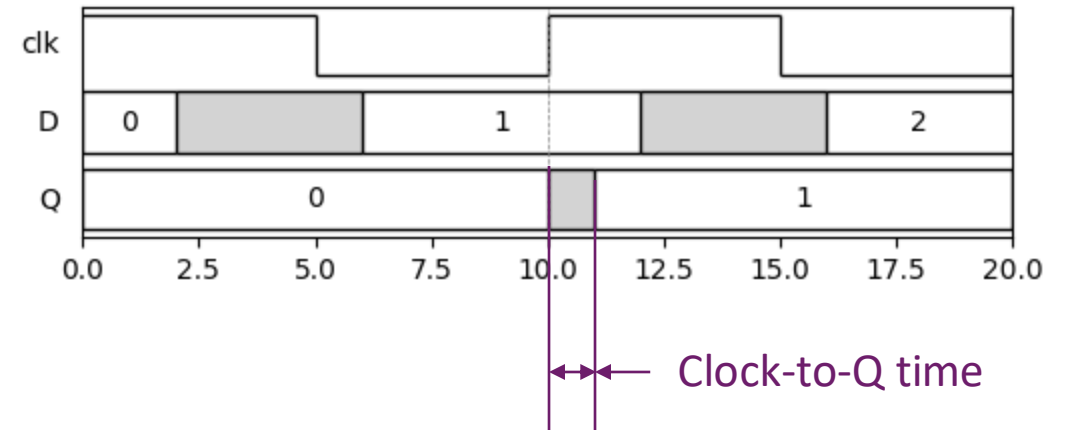
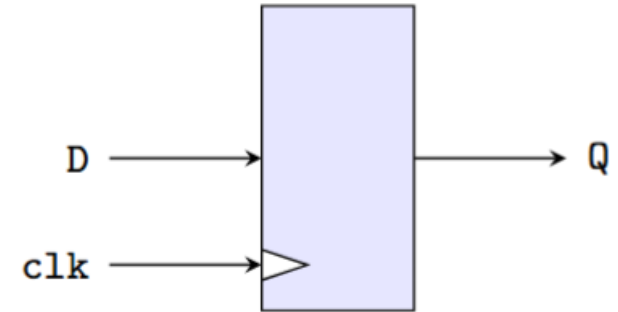
Clock-to-Q time

□ Clock-to-Q time

- Time from rising edge to stable output Q

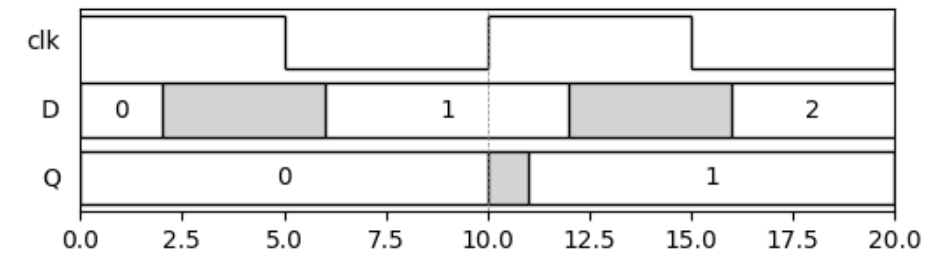
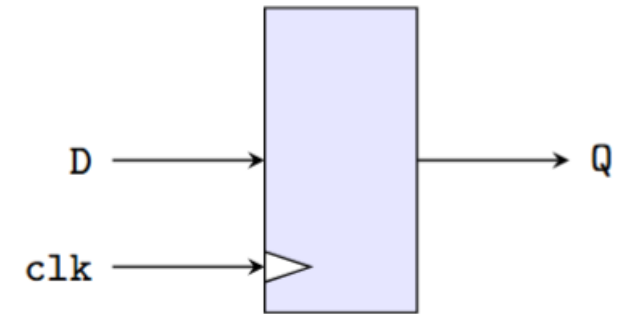
□ Sometimes loosely called register delay

□ Physically time to transfer input to output



Typical Times

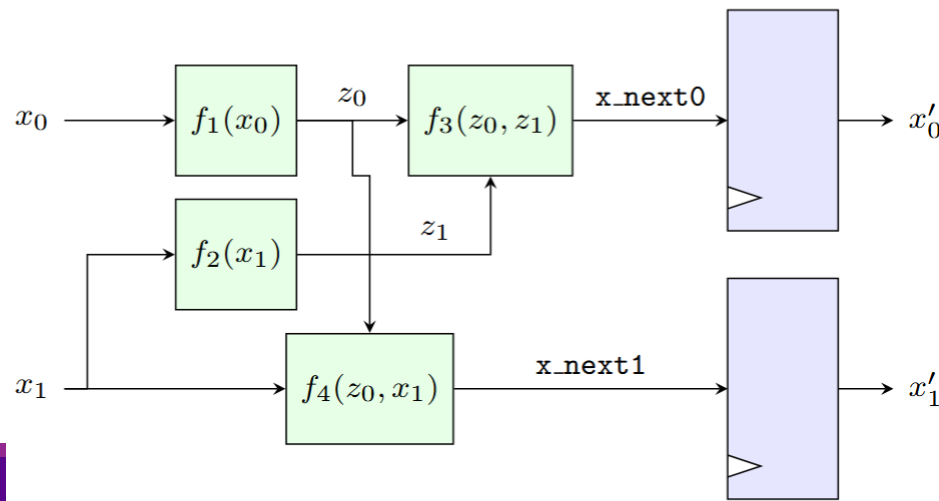
Technology	Clock-to-Q	Setup Time	Hold Time	Typical Clock Period
Low-end FPGA (e.g., Pynq / Zynq-7000 / Artix-7)	300–600 ps	150–300 ps	0–150 ps	5–10 ns (100–200 MHz)
High-end FPGA (e.g., Xilinx UltraScale+, Intel Stratix 10)**	150–300 ps	100–200 ps	0–120 ps	1.5–3 ns (300–650 MHz)
ASIC (modern nodes, e.g., 7–28 nm)	20–80 ps	20–60 ps	–20 to +40 ps	0.3–1.0 ns (1–3+ GHz)



Note: Modern nodes may have a negative hold time!

Register-to-Register Paths

```
always_comb begin
    z0 = func1(x0);
    z1 = func2(x1);
    x_next0 = func3(z0, z1);
    x_next1 = func4(z0, x1);
end
always_ff @(posedge clk) begin
    x0 <= x_next0;
    x1 <= x_next1;
end
```



□ Register-to-register paths

- Start at any register output
- End at any register output

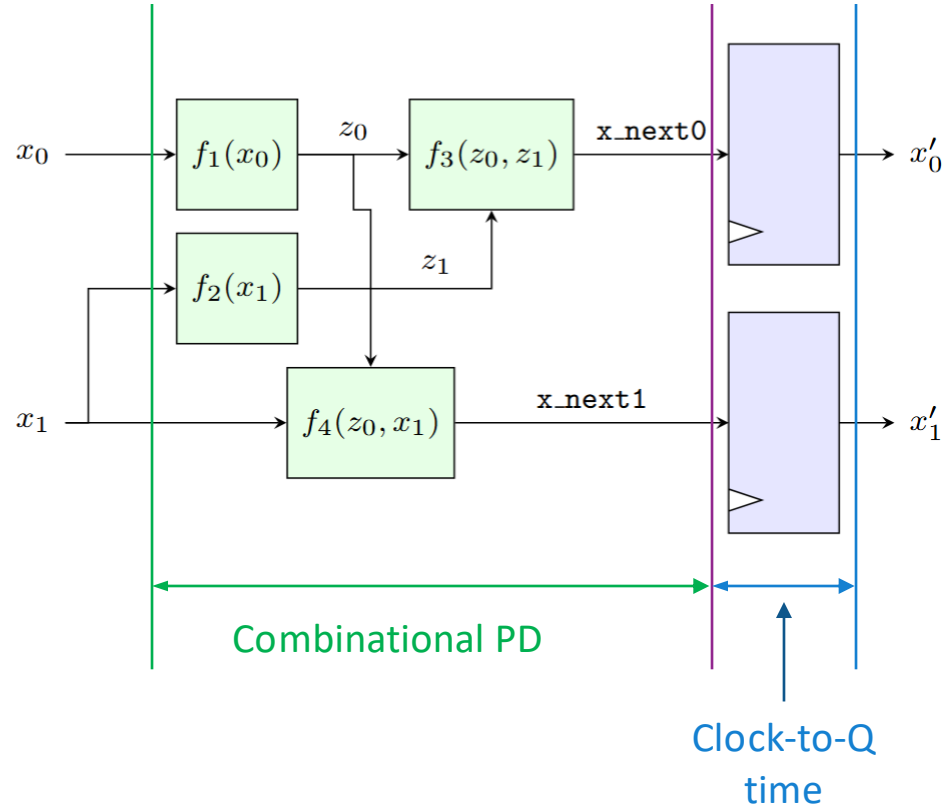
□ Can be visualized on a dataflow graph

- Maps logics and register for each update
- Primed values x'_0, x'_1 represent values in *next* cycle

□ In this example, there are four paths

- $x_0 \rightarrow f_1 \rightarrow f_3 \rightarrow x'_0$
- $x_0 \rightarrow f_1 \rightarrow f_4 \rightarrow x'_1$
- $x_1 \rightarrow f_2 \rightarrow f_3 \rightarrow x'_0$
- $x_1 \rightarrow f_2 \rightarrow f_4 \rightarrow x'_1$

STA Analysis: Verifying the Set-Up Time



- ❑ For now: Assume min hold time condition is met
 - We will address this later
- ❑ Must verify the set-up time condition:
 - Max path delay $\leq T_{clk} - T_{setup}$
 - All paths arrive within set up period before clock edge
- ❑ Path delay has two components
 - T_{PD} = Combinational propagation delay
 - T_{clk-Q} = One clock-to-Q delay
- ❑ Now max $T_{PD} = T_{crit}$ = Max PD of critical path
 - Measured in combinational circuit
- ❑ Hence, to meet timing we need:
$$T_{crit} + T_{clk-Q} + T_{setup} \leq T_{clk}$$

Meeting or Failing Timing

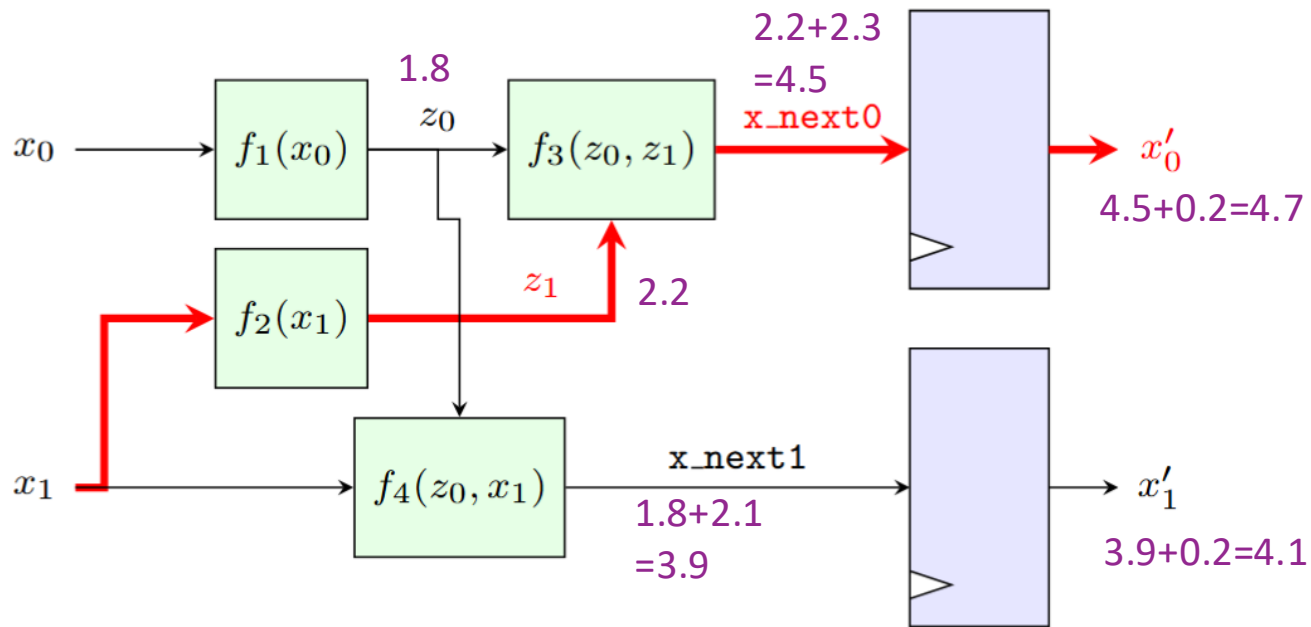
- From previous slide, assuming minimum hold time condition is met:

$$\text{To meet timing: } T_{clk} > T_{crit} + T_{setup} + T_{CLK-Q}$$

- T_{clk} = clock period, $f_{clk} = \frac{1}{T_{clk}}$ = clock frequency
 - T_{crit} = delay of critical path in combinational logic
 - T_{setup} = min set up time, T_{CLK-Q} = Clock-to-Q time
- If condition is satisfied, say system **meets timing**
 - Otherwise, system **fails timing / timing violation**

Example: Calculating Critical Path

- Max propagation delay
 - Can be computed recursively
 - Identical to combinational section



Component	Value [ns]
f_1 PD	1.8
f_2 PD	2.2
f_3 PD	2.1
f_4 PD	2.3
T_{clk-Q}	0.2
T_{setup}	0.1

Example: Maximum Clock Frequency

Component	Value [ns]
T_{crit}	4.5
T_{clk-Q}	0.2
T_{setup}	0.1
$T_{clk \text{ min}}$	4.8

□ From previous slide:

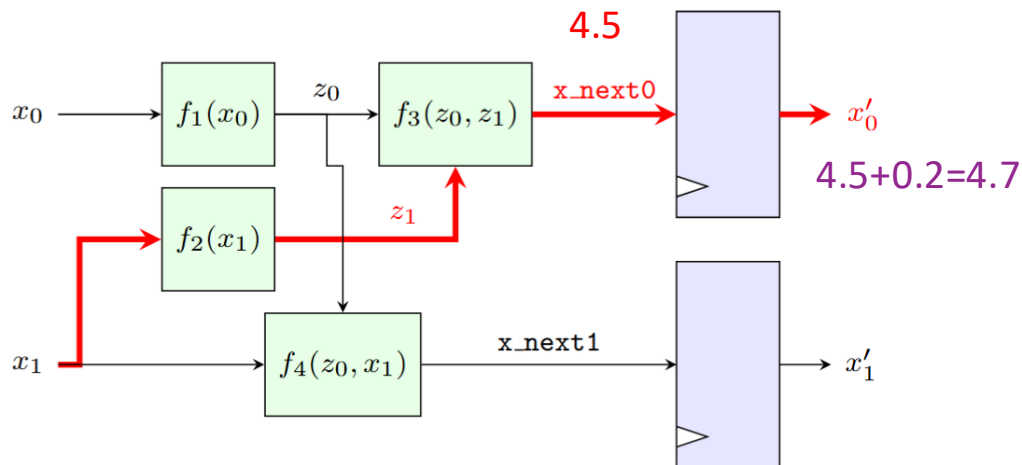
- Critical path delay in combination logic: $T_{crit} = 4.5$ ns

□ Therefore, to meet timing:

$$T_{clk} \geq T_{crit} + T_{CLK-Q} + T_{setup} \\ = 4.5 + 0.2 + 0.1 = 4.8 \text{ ns}$$

□ Max clock frequency

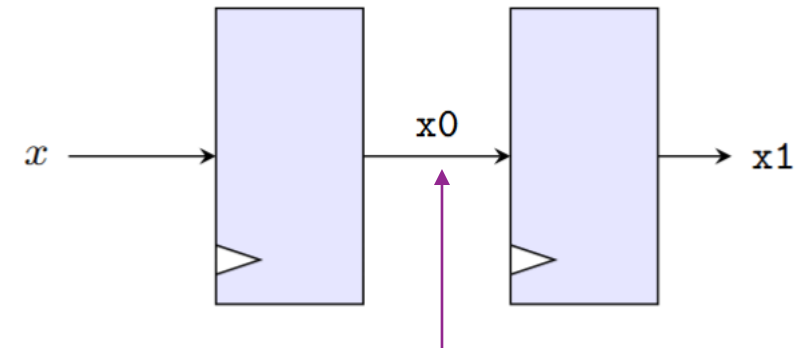
$$f_{clk} = \frac{1}{T_{clk}} \leq 208 \text{ MHz}$$



Meeting the Min Hold Time Constraint

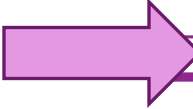
- ❑ Consider circuit to the right
 - A simple delay line
- ❑ Naïve implementation
 - Very short path: Output $x_0 \rightarrow$ input x_1
 - Could violate minimum hold time
- ❑ Typical solved in place & route
 - Physical design will insert delay elements
- ❑ With reasonable synchronous design:
 - RTL design does not need to worry about this constraint

```
always_ff @(posedge clk) begin
    x0 <= x;
    x1 <= x0;
end
```

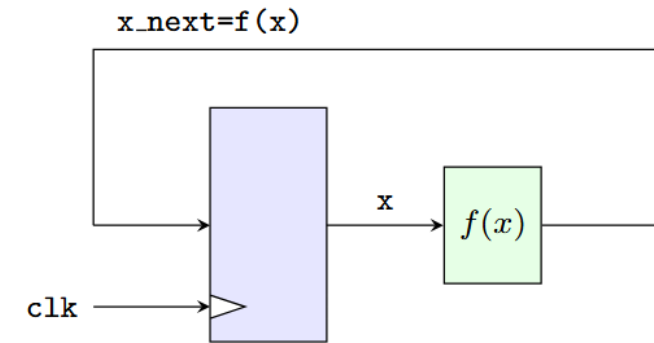
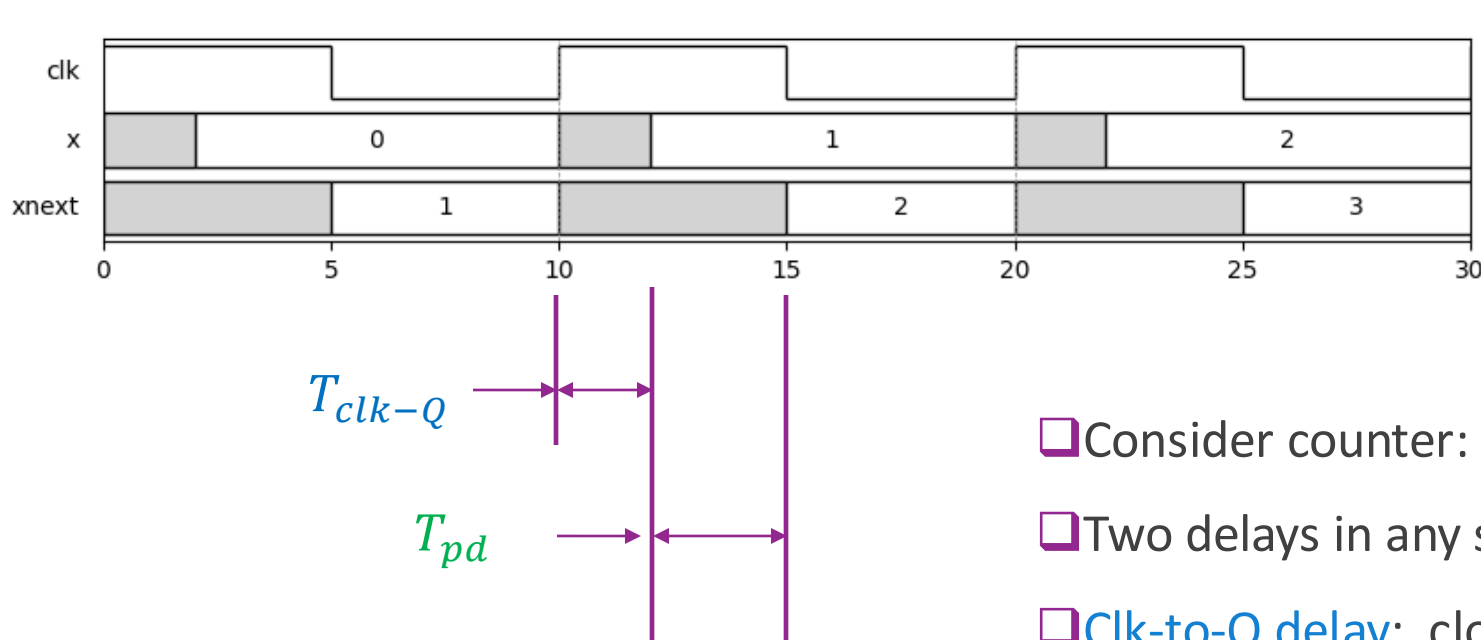


Short path.
Could violate min hold

Outline

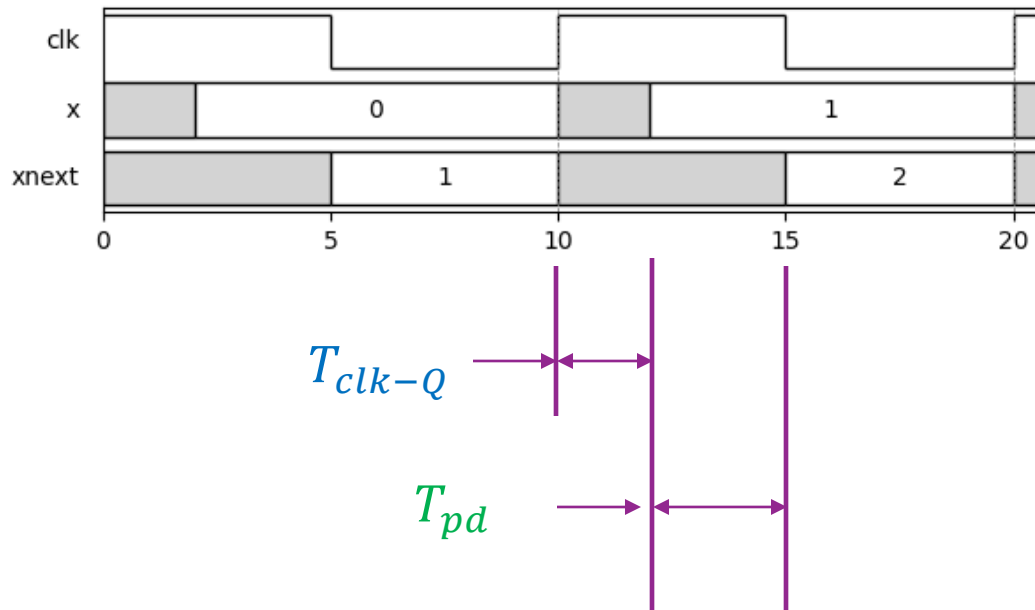
- ❑ Static Timing Analysis in Combinational Logic
- ❑ Static Timing Analysis in Sequential Logic
-  Sequential Logic Timing Diagrams
- ❑ VCD Files and Timing Diagrams from Vitis

Synchronous Logic Detailed Timing Diagram



- ❑ Consider counter: $f(x) = x + 1$
- ❑ Two delays in any synchronous circuit
- ❑ **Clk-to-Q delay**: clock rising edge $\rightarrow$ register output
 - This example shown as 2 ns
- ❑ **Combinational PD**: register output $\rightarrow$ next value
 - This example shown as 3 ns

Detailed Timing Diagram



❑ Timing diagram drawn called **detailed diagram**

- Sometimes called the **real timing diagram**

❑ Shows all details:

- Register delays
- All propagation delays in combinational paths

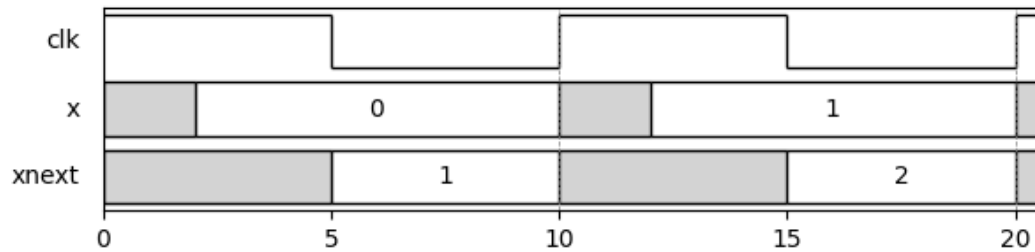
❑ Good for reasoning about precise **timing**

- Can compute computation speed, throughput, ...

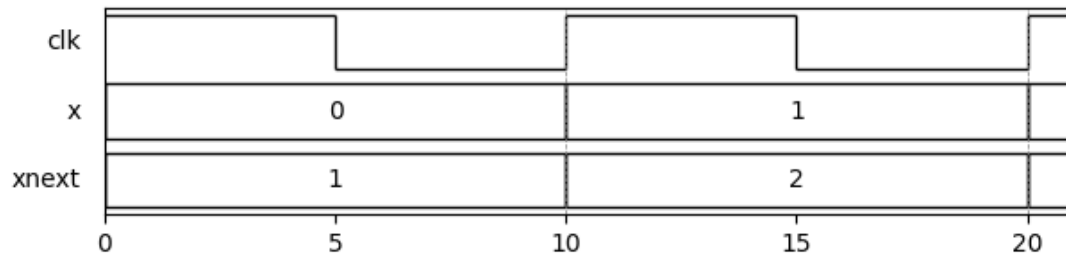
❑ But too many details to understand **functionality**

Abstract or Cycle Accurate Timing Diagram

Detailed timing diagram



Delays $\rightarrow 0$



Abstract timing diagram

□ Abstract timing diagram

- Also called **cycle-accurate diagram**
- All delays $\rightarrow 0$
- Ignores computation time

□ No unknown (X or ?) regions

□ Simpler to see functionality

□ Values in each cycle represent

- Value immediately before rising edge at end of cycle

ReLU + Linear Synchronous Module

- Now consider synchronous version of ReLU + linear
- Identical to Unit 2
- We add two inputs to ports
 - Clock signal
 - Reset on negative edge

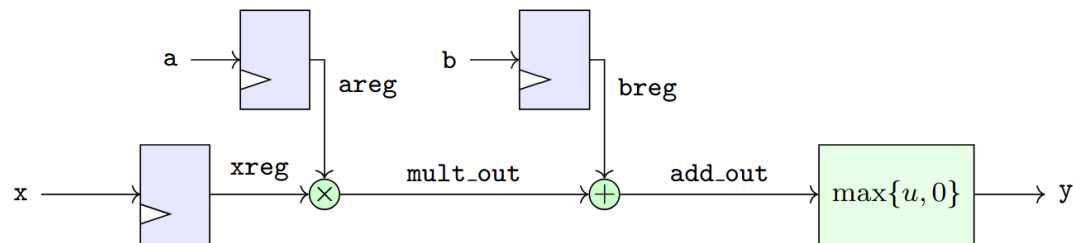
```
module relu_lin (  
    input logic clk,   
    input logic rst_n,   
    input logic int x,   
    input logic int a,   
    input logic int b,   
    output logic int y  
);
```

← Clock

← Reset

Synchronous Implementation

- ❑ Use identical implementation as in Unit 2
- ❑ Next values computed in always_comb block



```
int x_reg, a_reg, b_reg;

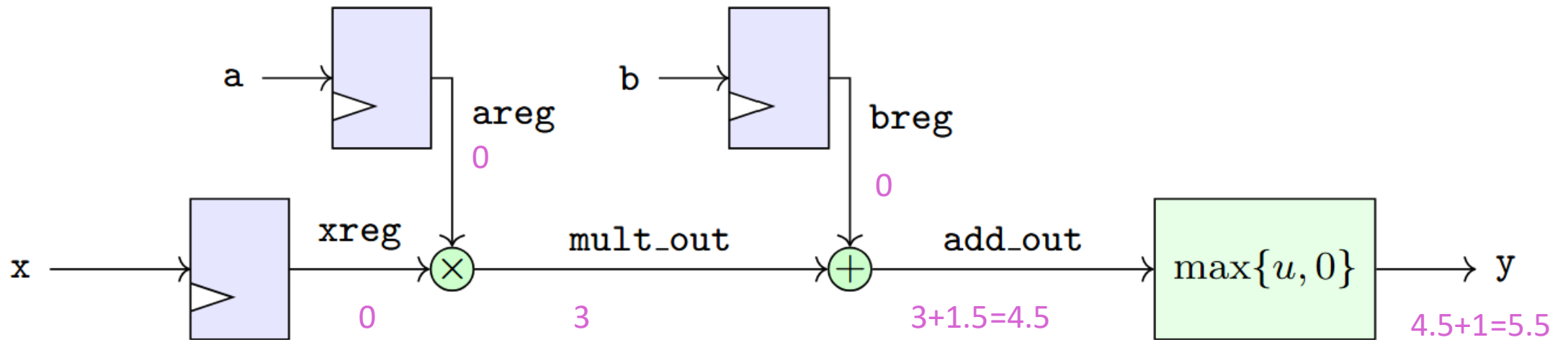
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        x_reg <= 0;
        a_reg <= 0;
        b_reg <= 0;
    end else begin
        x_reg <= x;
        a_reg <= a;
        b_reg <= b;
    end
end

always_comb begin
    logic int mult_out, add_out;
    mult_out = x_reg * a_reg;
    add_out = mult_out + b_reg;
    y = (add_out > 0) ? add_out : 0;
end
endmodule
```

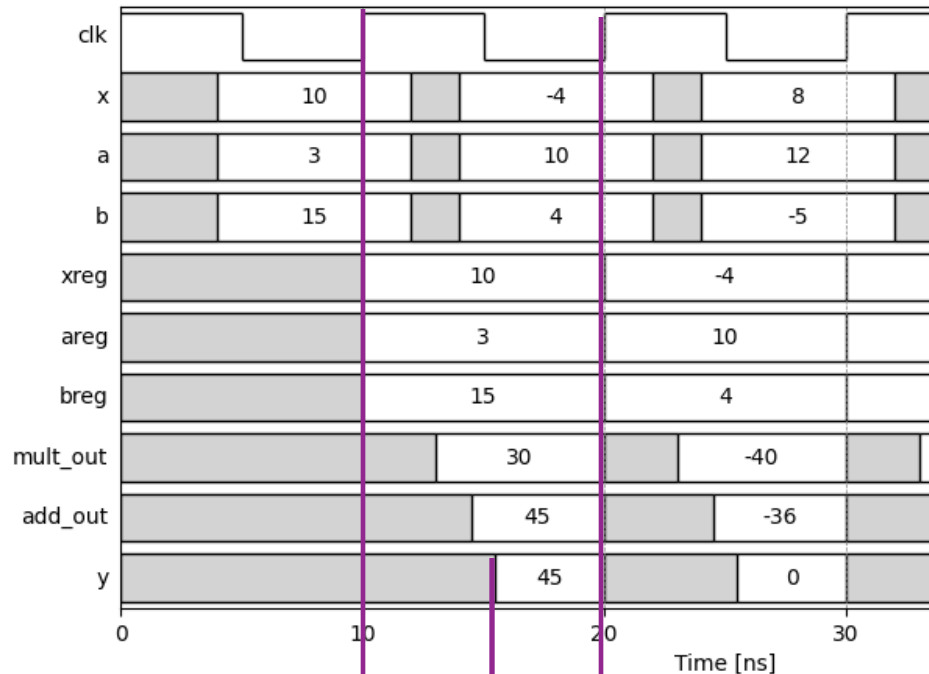
STA

- Assume the propagation delays in table
- Delay in each stage shown in violet
- Ignore CLK-Q time

Hardware block	Max prop delay
Register	0 ns
Multiplier	$T_{mult} = 3$ ns
Adder	$T_{add} = 1.5$ ns
Max operator	$T_{max} = 1$ ns
Total =critical path	$T_{crit} = 5.5$ ns



ReLU+Linear Timing Diagram

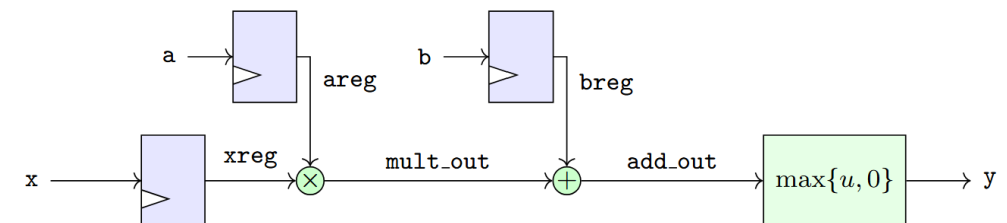


$$T_{crit} = 5.5 \text{ ns}$$

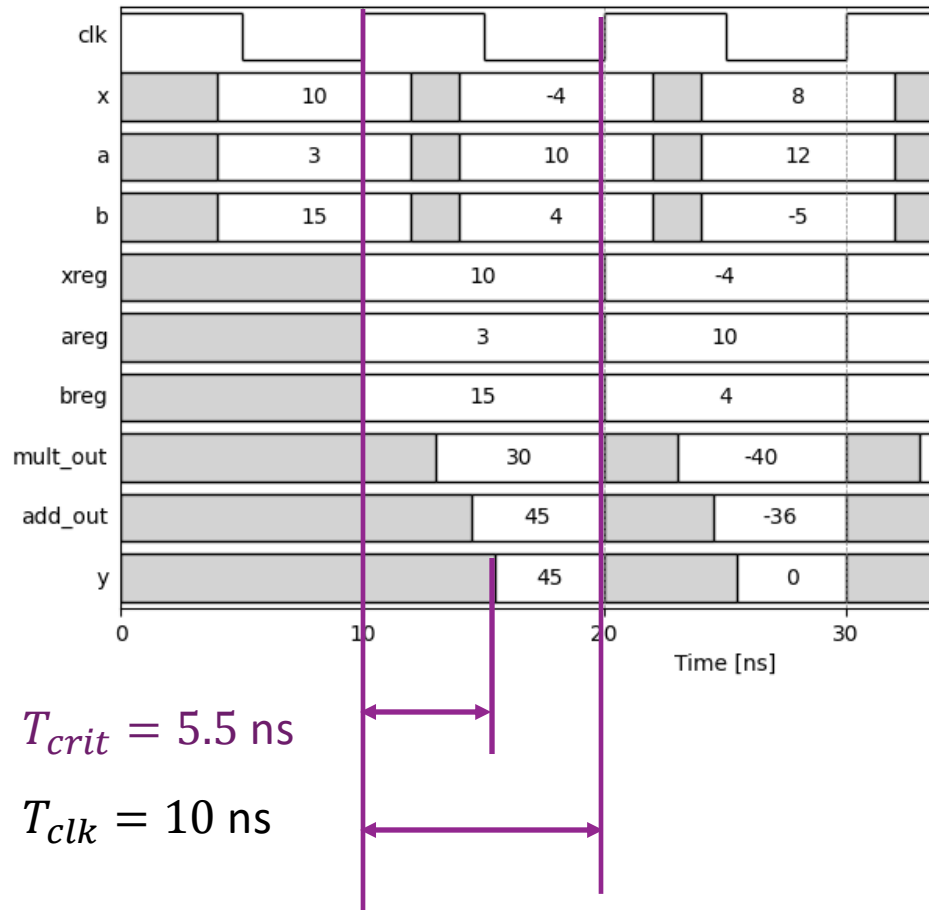
$$T_{clk} = 10 \text{ ns}$$

Meets timing:
 $T_{clk} > T_{crit}$

Hardware block	Example Value
Multiplier	$T_{mult} = 3 \text{ ns}$
Adder	$T_{add} = 1.5 \text{ ns}$
Max operator	$T_{max} = 1 \text{ ns}$
Total =critical path	$T_{crit} = 5.5 \text{ ns}$

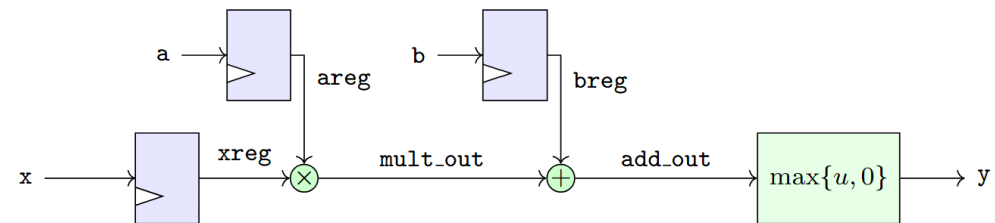


Max Clock Frequency Example



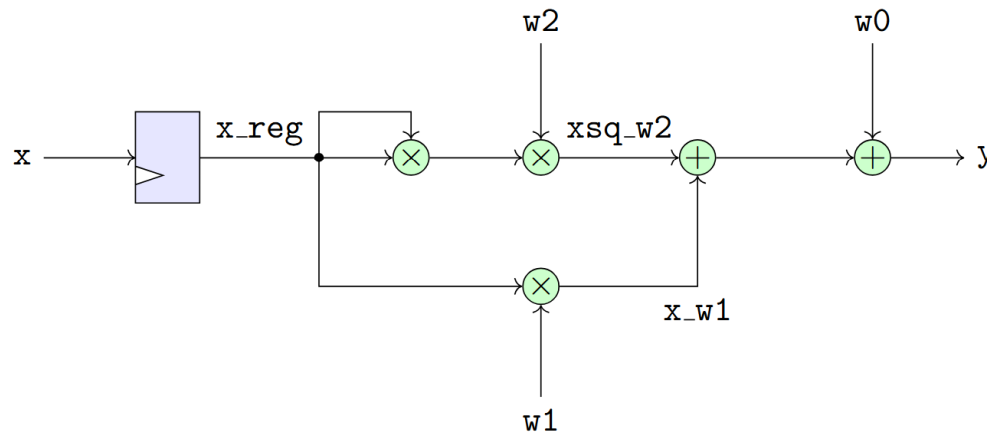
□ Max clock frequency in our example:

- $\text{Min } T_{clk} = T_{crit} = 5.5 \text{ ns}$
- $\text{Max clock frequency} = \frac{1}{5.5} \cong 180 \text{ MHz}$



Quadratic Example

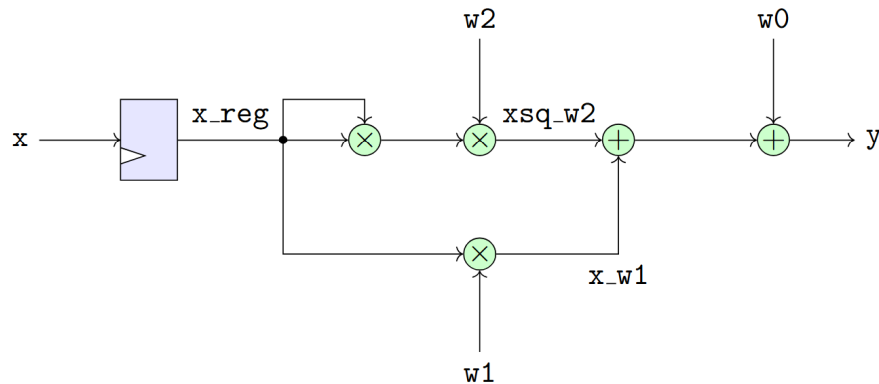
- ❑ Consider a quadratic function: $y = w_2x^2 + w_1x + w_0$
- ❑ For simplicity, assume w_0, w_1, w_2 are fixed parameters
- ❑ Single cycle implementation to the right



```
module quad_func #(
    parameter int w2 = 0,
    parameter int w1 = 0,
    parameter int w0 = 0
) (
    input logic clk,
    input logic int x,
    output logic int y
);
    int xreg;
    always_ff @(posedge clk) begin
        xreg <= x;
    end
    always_comb begin
        logic int xsq, xsq_w2, x_w1;
        xsq = xreg * xreg;
        xsq_w2 = w2 * xsq;
        x_w1 = w1 * xreg;
        y = xsq_w2 + x_w1 + w0;
    end
end
```

Finding the Critical Path

Hardware block	Example Value
Register	$T_{reg} = 1 \text{ ns}$
Multiplier	$T_{mult} = 4 \text{ ns}$
Adder	$T_{add} = 1.5 \text{ ns}$
Path 1	$1+4+4+1.5+1.5 = 12$
Path 2	$1+4+1.5+1.5=8$



❑ Consider implementation with times in table

❑ Two paths:

❑ Path 1: x, x_reg, xsq_w2, y

- 1 register, 2 mults, 2 adders = 12 ns

❑ Path 2: x, x_reg, x_w1, y

- 1 register, 1 mult, 2 adders = 8 ns

❑ Path 1 is critical

❑ Max clock frequency = $\frac{1}{12 \text{ ns}} = 83 \text{ MHz}$

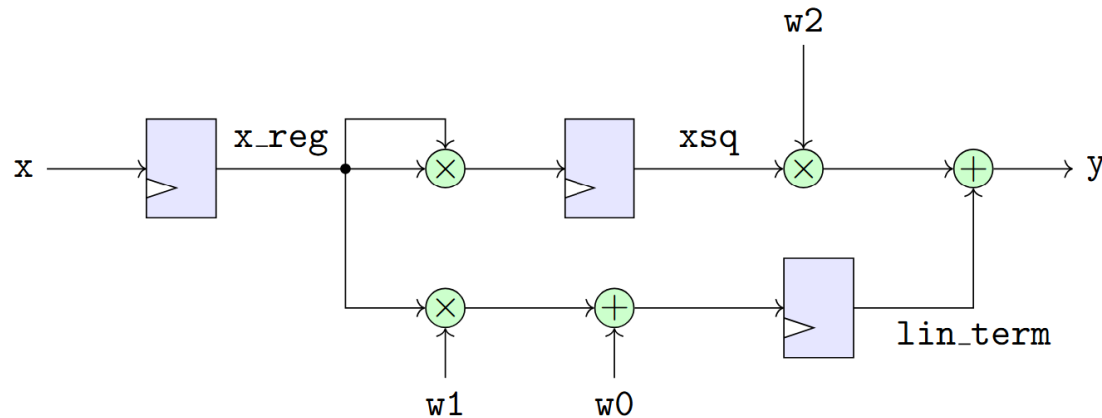
❑ Would **fail timing** at 100 MHz

A Two Cycle Design

❑ We can improve throughput by breaking computation over two cycles

❑ Register two intermediate terms:

- $xsq = xreg * xreg$
- $lin_term = w1 * xreg + w0$



```
int xreg;
always_ff @(posedge clk) begin
    xsq <= xreg * xreg;
    lin_term <= w1 * xreg + w0;
    xreg <= x;
end
always_comb begin
    y = w2*xsq + lin_term;
end
```

Three Stages

❑ Design is processed in three stages = 2 cycles

❑ Stage 0

- Register input x to `xreg`

❑ Stage 1: Compute and register:

- $xsq = xreg * xreg$
- $lin_term = w1 * xreg + w0$

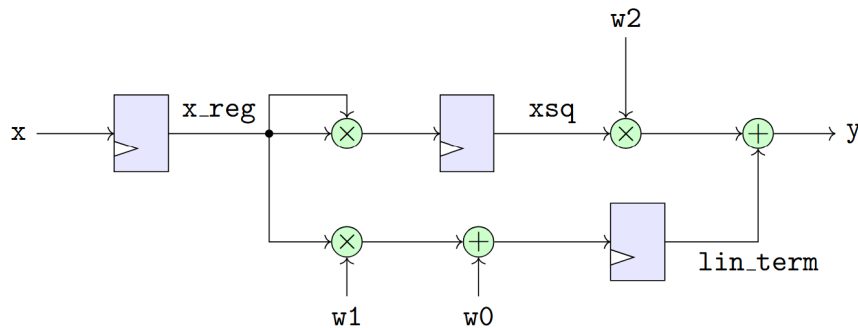
❑ Stage 2: Compute output

- $y = w2 * xsq + lin_term$

```
int xreg;
always_ff @(posedge clk) begin
    xsq <= xreg * xreg;
    lin_term <= w1 * xreg + w0;
    xreg <= x;
end
always_comb begin
    y = w2*xsq + lin_term;
end
```

Critical Path with Three Stages

Hardware block	Example Value
Register	$T_{reg} = 1 \text{ ns}$
Multiplier	$T_{mult} = 4 \text{ ns}$
Adder	$T_{add} = 1.5 \text{ ns}$
Path 1	$1+4=5$
Path 2	$1+4+1.5=6.5$
Path 3	$1+4+1.5=6.5$



❑ Measure paths from register to register

❑ Path 1: x, x_reg, xsq input

- 1 register, 1 mult = 5 ns

❑ Path 2: $x, x_reg, w1_x, lin_x$ input

- 1 register, 1 mult, 1 add = 6.5 ns

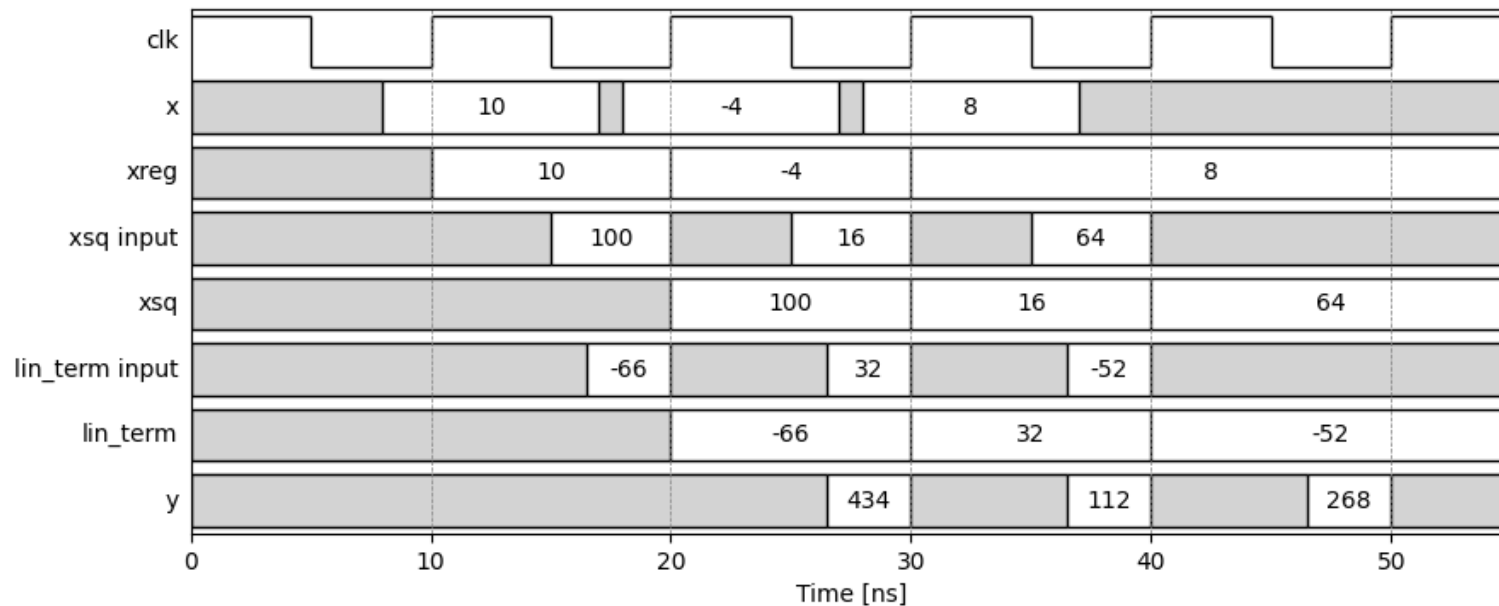
❑ Path 3: $xsq_reg, lin_x, w2_xsq, y$

- 1 register, 1 mult, 1 add = 6.5 ns

❑ Critical path delay = 6.5 ns

❑ Max freq = $\frac{1}{6.5} \approx 153 \text{ MHz}$

Two Cycle Timing Diagram



Example with

$w = [4, -7, 5]$

x inputs 10, -4, 8

Latency, Throughput and Pipelining

□ Suppose that we run at our design at 100 MHz clock

□ Latency:

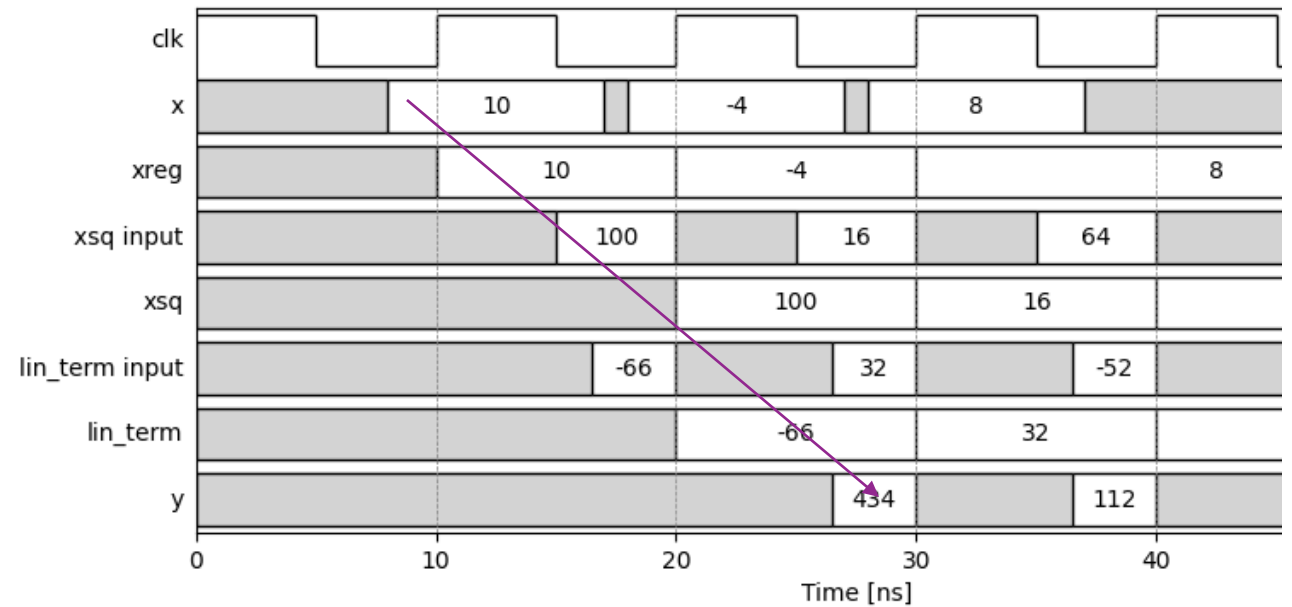
- Suppose input is valid on rising edge of clock n
- Then output is valid before rising edge of clock $n + 2$
- Latency = 2 clock cycles = 20 ns

□ Throughput:

- Circuit can take a new input each cycle
- Throughput = 1 input / cycle = 100 MHz

□ Process is called a pipeline

- Circuit is processing multiple inputs at the same time

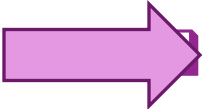


Performance Comparison

	Single Cycle Design	Two Cycle Design
Critical path	12 ns	6.5 ns
Max clock frequency	≈ 80 MHz	≈ 150 MHz
Max throughput	80 MHz	150 MHz
Latency	1 cycle = 12 ns	2 cycles = 13 ns

*Pipelining benefit:
For small increase in latency, we dramatically improve throughput*

Outline

- ❑ Static Timing Analysis in Combinational Logic
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- ❑ Sequential Logic Timing Diagrams
-  VCD Files and Timing Diagrams from Vitis

Value-Change Points

- ❑ Simulation tools trace values over time for large numbers of signals
- ❑ Data on each signal is typically stored via **value-change points**

Definition: A signal $x(t)$ has **value-change points** $\{(t_i, x_i), i = 0, \dots, n - 1\}$ if:

$$x(t) = \begin{cases} X & t < t_0 \\ x_i & t_i \leq t < t_{i+1}, \quad i = 0, \dots, n - 2 \\ x_{n-1} & t \geq t_{n-1} \end{cases}$$

- Value change points also called **valued change events, transitions, ...**

Timing Diagram and VCDs

□ Timing diagrams can be drawn from the VCDs

□ Example with two signals:

- VC points for x : $\{(3,5), (20,17), (40,8)\}$
- VC points for y : $\{(12,3), (25,12), (35,0)\}$

□ In this plot, times are in ns



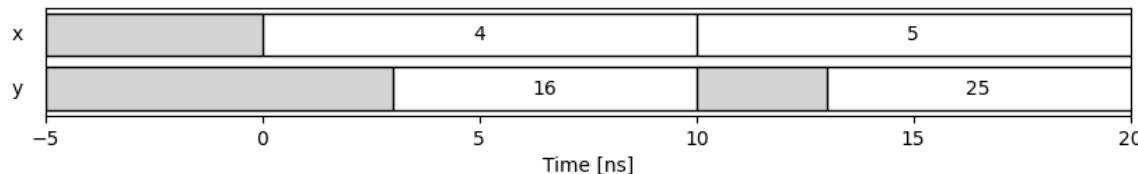
Drawing Timing Diagrams in Python

❑ Course repo contains a [xilinxutils package](#)

- Custom package in this class
- Many helper routines for Xilinx tools

❑ Tools for parsing and visualizing VCD files

❑ Code to produce timing diagram



[Hardware Design](#) / [Support Material](#) / [GitHub Repository](#) / [Inst](#)

Installing `xilinxutils`

Installing the package

```
xvals = ['x', '5', '17', '8']
yvals = ['x', '3', '12', '0']
xtimes = [0, 3, 20, 40]
ytimes = [0, 12, 25, 35]
```

```
xsig = SigTimingInfo("x", xtimes, xvals)
ysig = SigTimingInfo("y", ytimes, yvals)
td = TimingDiagram()
td.add_signals([xsig, ysig])
ax = td.plot_signals(trange=[0,50])
_ = ax.set_xlabel("Time [ns]")
```